Multiscale Split-Recombination Cooperative Fusion Network for Hyperspectral and LiDAR Land Cover Classification

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***Abstract*— In recent years, with the continuous advancement of Earth observation technologies, the joint utilization of hyperspectral imagery (HSI) and light detection and ranging (LiDAR) for classifying complex land cover types has garnered widespread attention in the field of multisource remote sensing. However, the diversity of land cover increases the complexity of the spatial and spectral structures in remote sensing data, creating challenges for extracting discriminative features effectively. Moreover, the heterogeneity of multisource remote sensing data poses challenges for existing methods to effectively integrate complementary information from various data sources. To address these challenges, a multiscale split-recombination cooperative fusion network (MSRCFNet) is proposed for joint land cover classification using HSI and LiDAR data. It consists of three main components: multiple parallel multiscale hierarchical inverted-pyramid (MHIP) modules, a cross-modal cooperative fusion module (CCFM), and a classification module. The MHIP module comprises multiple convolutional split-recombination blocks (CSRBs) at various scales, designed to extract and fuse discriminative multiscale features. CCFM leverages spatial-scale consistency and the self-attention mechanism to model global relationships between different modalities, and then applies the cross-attention mechanism to effectively integrate complementary information from heterogeneous data. The classification module transforms the fused features into the final classification results. Experimental results on three publicly available HSI-LiDAR datasets demonstrate the superiority of the proposed network over state-of-the-art methods.**

***Index Terms*— Attention mechanism, cooperative fusion, multiscale feature extraction, multisource remote sensing.**

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I. Introduction

As REMOTE sensing technology advances, remote sensing imagery applications in Earth observation emerge as a prominent research focus [1]. Using sensors to acquire remote sensing imagery for land cover identification represents a significant approach to addressing complex surface classification challenges [2], [3]. The abundant multisource remote sensing data not only contributes significantly to Earth observation but also provides new methods and presents challenges for the broad application of remote sensing across various fields [4], [5]. Hyperspectral imagery (HSI) has emerged as a focal point in remote sensing research, receiving growing attention due to its capacity to capture numerous continuous spectral bands, providing both detailed spectral and spatial information [6]. Due to its distinct properties, HSI finds widespread application across various fields, including environmental monitoring [7], mineral analysis [8], and precision agriculture [9]. However, as a passive remote sensing technology, HSI is susceptible to interference from atmospheric conditions and lighting, leading to variations in spectral bands in real-world scenarios [10]. This characteristic poses numerous challenges when applying HSI to complex scenes. Furthermore, a single remote sensing data can only capture specific attributes and fails to effectively represent the complete information of the scene [11].

To address the aforementioned issues, researchers extensively investigated combining data from multiple remote sensing sources over the past decade [12]. Light detection and ranging (LiDAR) is an active sensing technology that accurately captures the spatial structure and elevation data of objects on the ground [13]. In addition, compared with HSI, LiDAR is less affected by factors such as lighting conditions, weather changes, and the reflectivity of surface materials, showing strong environmental adaptability. However, under extreme weather conditions, such as thunderstorm weather, it may cause the laser signal to scatter and introduce measurement noise [14]. Due to differences in sensor types, multisource remote sensing data are classified into homogeneous and heterogeneous data [15]. HSI and LiDAR represent two types of heterogeneous data, each containing spatial–spectral information and spatial-elevation information, respectively. Research shows that for surface cover types with similar spectral curves but different elevations,

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LiDAR-derived elevation data effectively alleviates the same-spectrum-different-objects phenomenon in HSI data [16]. However, LiDAR has limited capability in addressing the different-spectrum-same-objects phenomenon, which primarily requires spectral resolution improvement. Therefore, the HSI and the LiDAR data mutually complement one another in land cover classification tasks, which significantly enhances the classification accuracy for land use and land cover (LULC) applications.

In early studies on the joint classification of HSI and LiDAR data, traditional machine learning algorithms serve as key methods for researchers in addressing the problem. Due to the complexity of the observed targets, feature-level and decision-level fusion methods are commonly used to integrate data [17]. Dalponte et al. [18] classify complex forest regions by concatenating HSI and LiDAR data, using support vector machines (SVMs) and Gaussian maximum likelihood methods. However, this fusion approach leads to the stacking of multiple features, thus increasing data dimensionality. The high-dimensional nature of the data increases processing complexity and reduces computational efficiency. To address this issue, researchers propose dimensionality reduction techniques such as principal component analysis (PCA) [19] and linear discriminant analysis (LDA) [20]. These techniques not only reduce data dimensionality but also effectively minimize redundant spectral information in HSI. Van Coillie et al. [21] apply PCA to process the fused HSI and LiDAR data, significantly improving accuracy on small sample datasets. Compared to feature-level fusion, decision-level fusion involves classifying different remote sensing data independently and then integrating the classification results. This fusion strategy effectively avoids the direct handling of high-dimensional data, making it suitable for scenarios involving large datasets. Bigdeli et al. [22] separately train hyperspectral features and LiDAR features using SVM, and then apply the naive bayes method for decision-level fusion of the two classification results. Experimental results show that this fusion strategy effectively improves classification accuracy. However, traditional machine learning methods cannot independently extract deep features from multimodal data, which makes it difficult to capture the intricate nonlinear connections between HSI and LiDAR data. In addition, these approaches are prone to noise and outliers, limiting their effectiveness in handling large-scale, complex data.

Deep learning gains widespread attention in the field of HSI-LiDAR multisource data fusion classification because it fully leverages abundant computational resources and automatically learns deeper features from data with minimal human intervention [23]. Convolutional neural networks (CNNs) are among the earliest methods used to classify multisource remote sensing data jointly [24]. The different modal features are individually extracted using the extinction profiling (EP) technique before being fed into the CNN to generate the classification results. Many CNN-based methods employ 2D-CNNs to extract local texture features from pixel-centered data cubes. Zhao and Du [25] use a locally discriminative embedding method to extract features from high-dimensional HSI by employing a 2D-CNN. However, the dimensional limitations of 2D-CNN hinder its ability to fully capture the spectral features. To address this, Chen et al. [26] introduce a 3D-CNN-based regularized deep feature extraction method. This method employs multiple convolutional layers to extract deep features while utilizing L2 regularization and dropout techniques to mitigate challenges posed by limited training samples. The aforementioned methods are limited to extracting features from a single HSI. Roy et al. [27] utilize a 3D-CNN combined with spatial morphological techniques for feature learning on HSI and LiDAR fused data. This approach treats the elevation information from LiDAR as a complement to HSI, which effectively enhances robustness. Zhang et al. [28] employ a bidirectional autoencoder to reconstruct heterogeneous data and feeds the fused multisource data into two CNN branches for final classification. However, because of the scattered nature of various land cover types, extracting single-scale features alone does not effectively capture the spatial relationships among feature types. To address this, many researchers introduce multiscale approaches to extract more detailed and comprehensive features. Pan et al. [29] introduce a multiscale cross-fusion network, which utilizes different multiscale feature extraction modules to extract spectral–spatial features. The network employs distinct fusion strategies to effectively integrate homogeneous and heterogeneous data. Fan et al. [30] adopt multiscale self-calibrated convolutions to capture information across different scales and apply a hierarchical residual structure to improve the features extracted at different scales, thereby improving feature representation capabilities. Gao et al. [31] employ a multibranch hierarchical fusion strategy to extract multiscale spectral–spatial features. In addition, the model uses three convolutional layers to extract shallow elevation features, providing complementary information to the HSI features. Wang et al. [32] design a multiscale pyramid fusion framework to extract multiscale features.

CNNs play a crucial role in feature extraction but cannot capture the correlation information between features. In addition, due to the limitations of the receptive field, CNNs capture only local features and often overlook long-range dependencies between spectral features. With the widespread application of transformers and attention mechanisms in fields like natural language processing (NLP) [33], new ideas and methods emerge for researchers. Dosovitskiy [34] propose a model called vision transformer (ViT), which has a significant impact on the field of computer vision. This study demonstrates that, without relying on CNNs, using transformers directly for image classification can still achieve satisfactory results. Transformers, in contrast to CNNs, utilize the inherent structure of the data to capture long-range dependencies globally [35]. This flexible approach to data integration effectively drives the rapid development of the HSI-LiDAR multisource fusion classification field. Xue et al. [36] propose a deep hierarchical ViT architecture that uses self-attention to capture long-range dependencies and cross-attention to fuse heterogeneous features adaptively. Roy et al. [37] introduce a novel cross-hyperspectral and LiDAR attention mechanism based on ViT, which leverages pixel-level correlations in remote sensing scenes to fuse heterogeneous data, effectively

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improving classification performance. Zhang et al. [38] propose a local information interaction transformer, which designs a multisource feature selection module to fuse multisource data. Hong et al. [39] develop a multimodal transformer network, which captures both unique and common features from multimodal data. Due to the differences and complementary nature of CNNs and transformers in data processing, many methods combining the two have emerged. Zhao et al. [40] propose a method that integrates CNNs and transformers to fully fuse multisource heterogeneous data, incorporating a cross-fusion encoder to effectively combine the heterogeneous data. Sun et al. [41] propose a mixed CNN and lightweight attention-free transformer network, which feeds multimodal features extracted by hybrid convolutions into an attention-free encoder. This approach reduces training parameters while maintaining good classification performance. Wang et al. [42] develop a hybrid convolution module based on 3D-CNN and a transformer-based cross-modal enhanced deep feature fusion module. Wang et al. [43] introduce a convolution interaction transformer network, which uses asymmetric depthwise convolution and transformer to construct a local–global feed forward network model, extracting and fusing local and global features from shallow to deep levels in multimodal data.

Although CNN-based and transformer-based methods have advanced HSI-LiDAR joint classification, several key issues still remain.

1) Insufficient multiscale information extraction leads to existing networks being unable to fully exploit both local detailed information and global contextual information in multisource remote sensing data, causing the correlation between local and global features to be overlooked.
2) Most current methods are limited to extracting and fusing shallow features and fail to fully utilize deep features to comprehensively represent the complex structure and semantic information of multisource data.
3) Although multisource data have strong complementarity, effectively integrating the complementary information between heterogeneous data remains a problem.

To address these issues, a multiscale split-recombination cooperative fusion network (MSRCFNet) is proposed. The proposed method consists of three parallel multiscale hierarchical inverted-pyramid (MHIP) modules, a cross-modal cooperative fusion module (CCFM), and a classification module. Specifically, a hierarchical-split fusion approach is adopted by the MHIP modules, with receptive fields at different scales used to extract multiscale features. A novel fusion strategy is employed to enhance the correlation between features at different scales. To integrate heterogeneous data, different attention mechanisms are employed by the CCFM to adaptively integrate the complementary features from various remote sensing data sources. Finally, a decision-level fusion strategy is applied to aggregate the fused features and produce the classification outputs. The key contributions of this study are outlined below.

1) To address the issue of insufficient multiscale feature extraction and enhance the correlation between heterogeneous features, the MSRCFNet is proposed. By extracting distinctive features across shallow and deep layers of multimodal data, MSRCFNet enables efficient fusion of these features, thus enhancing the ability of the model to capture comprehensive information and significantly improving classification performance.
2) To leverage deep features for representing the complex structures of multisource remote sensing data, an MHIP module is developed to capture and fuse multiscale features from both HSI and LiDAR data. Comprising several convolutional split-recombination blocks (CSRBs) at different scales, the MHIP module can extract multiscale features even from limited labeled samples. Through integrating features at various scales across adjacent branches, it strengthens the complementarity between local and global features. In addition, the multiple splitting and recombination operations within the CSRB enhance the module's capacity to process complex data, effectively addressing the deep-feature underutilization issue.
3) To achieve complementary advantages between heterogeneous data, a CCFM is designed for the deep feature fusion. The CCFM is composed of two stages. In the first stage, a multihead self-attention (MHSA) encoder is used to model long-range dependencies within each modality. In the second stage, a cross-modal cross-attention encoder is incorporated to achieve deep fusion between multimodal features. The CCFM adaptively assigns weights to different data sources, thus enhancing the overall classification performance.

The rest of this article is structured as follows. Section II reviews the related work. Section III presents the details of the proposed network. Section IV presents the experimental results and includes a discussion. Finally, Section V concludes the article and outlines future research directions.

II. Related Work

A. Multisource Remote Sensing Data Fusion Method

Unlike traditional methods of HSI classification, multisource remote sensing data addresses the challenges associated with relying on a single data source. However, capturing the correlations between different data sources remains a critical challenge. Morchhale et al. [44] propose a CNN-based pixel-level fusion method, successfully integrating the two modalities of data and addressing the impact of uncertainty factors. However, this fusion strategy primarily focuses on shallow feature fusion, which faces challenges in fully exploiting the correlation information between the two types of data. Compared to shallow feature fusion approaches, deep learning approaches perform feature-level and decision-level fusion, which significantly enhances the fusion effectiveness. Ge et al. [45] employ several residual networks to fuse the data, which effectively enhances the classification performance. Wang et al. [46] propose a Gaussian mixture variational dynamic fusion network, which enhances the adaptive fusion of multimodal data by constraining the topological consistency of the two modalities, thereby effectively suppressing spectral noise. Although these deep learning fusion

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methods address the limitations of traditional approaches in their insufficient integration of multimodal features, the issue of inadequate feature extraction remains. As a result, researchers improve deep learning algorithms. Zhao et al. [47] propose a hierarchical random walk network for feature extraction. Zhang et al. [48] propose a method that combines attention mechanisms with a CNN to enhance the extraction of features from multimodal data. These studies highlight the strengths of multimodal data fusion in enhancing feature representation and improving classification accuracy.

B. Multiscale Feature Extraction

Feature extraction aims to capture more relevant information to improve the training effectiveness of the network [49]. In multisource remote sensing data classification, extracting features at different scales provides more comprehensive information for image analysis due to the diversity and complexity of land cover. Small-scale features capture local detail information, while large-scale features provide global contextual information. Combining features of different scales addresses the problem of insufficient feature extraction with a single scale [50]. Currently, multiscale feature extraction methods have become a mainstream strategy for enhancing model performance. Ge et al. [51] design a network that combines pseudo-3-D convolution and multiscale pointwise convolution. The approach effectively extracts multimodal features and reduces computational complexity. Tong et al. [52] utilize the extended multiattribute profile to automatically extract structural information and apply multiscale total variation for feature extraction. Xi and Ye [53] use two different multiscale feature extraction methods to separately extract and fuse features from different modalities. These approaches prove effective in capturing both local details and global context features, thereby improving the accuracy and robustness of multisource data joint classification tasks.

C. Attention Mechansim

The attention mechanism [54] has been extensively researched and implemented in deep learning. Its core idea is to mimic the selective mechanism of human visual attention by dynamically assigning weights to emphasize the most important features of the input data relevant to the task [55]. Specifically, given input vectors for the query, key, and value, the output vector is derived as a weighted aggregation of the values. The weights for each value are set using a compatibility function that assesses the relationship between the query and its key. Fig. 1 illustrates the structures of the two attention mechanisms. The self-attention mechanism utilizes a shared input vector for the query, key, and value, which facilitates the modeling of global contextual relationships across elements at various positions [56]. The cross-attention mechanism [57] utilizes input vectors from multiple sources for the query, key, and value. It generates contextual representations by evaluating the relationships between the query and key, and facilitates the understanding of multisource feature relationships and the integration of information across sources. In recent years, attention mechanisms have been increasingly applied to areas including multimodal learning, time series analysis, and remote sensing classification tasks.

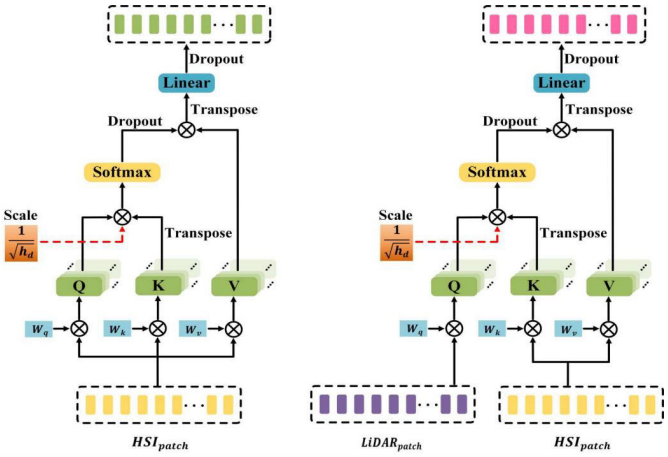


Fig. 1. Structures of the self-attention mechanism and the cross-attention mechanism.

D. Summary of Modeling Techniques

In this article, we employ several key modeling techniques. CNNs [58], [59], [60] are used to extract features from HSI and LiDAR data, taking advantage of their superiority in capturing spatial–spectral details. Inspired by human visual attention, the attention mechanism is integrated into the model to dynamically highlight key features, enhancing the ability of the model to recognize the relationships among multisource data [33]. In addition, cross-attention [61] is utilized to fuse heterogeneous features, leveraging their complementary information to improve the classification accuracy.

III. Methodology

This section provides the implementation details of the proposed MSRCFNet, with its detailed structural diagram presented in Fig. 2. The proposed network consists of three MHIP modules, a CCFM, and a final classification module. MSRCFNet utilizes a parallel approach for multiscale feature extraction, along with interactive integration of diverse features. Algorithm 1 presents the pseudo-code for training MSRCFNet for the fusion classification of HSI and LiDAR data.

Given an HSI denoted as $\boldsymbol{X}_H \in \mathbb{R}^{H\times W\times D}$ and a LiDAR image denoted as $\boldsymbol{X}_L \in \mathbb{R}^{H\times W}$, where H and W represent the image width and height, and D indicates the number of spectral bands of the HSI. To reduce the redundant spectral information in the HSI, PCA is applied for dimensionality reduction. The dimension-reduced HSI is denoted as $\boldsymbol{X}_H \in \mathbb{R}^{H\times W\times d}$. In order to make full use of the spatial information, the HSI is divided into pixel-centered $s \times s$ neighborhood cubes, and the LiDAR image is divided into pixel-centered $s \times s$ neighborhood patches. Each pixel of the HSI is extracted to obtain a small cube $\mathbf{X}_h^C \in \mathbb{R}^{s\times s\times d}$ through a 3-D block, while each pixel of the LiDAR image is extracted to obtain a small patch $\mathbf{X}_L^P \in \mathbb{R}^{s\times s}$ through a 2-D block. The central pixel index labels each patch, while edge pixels are padded

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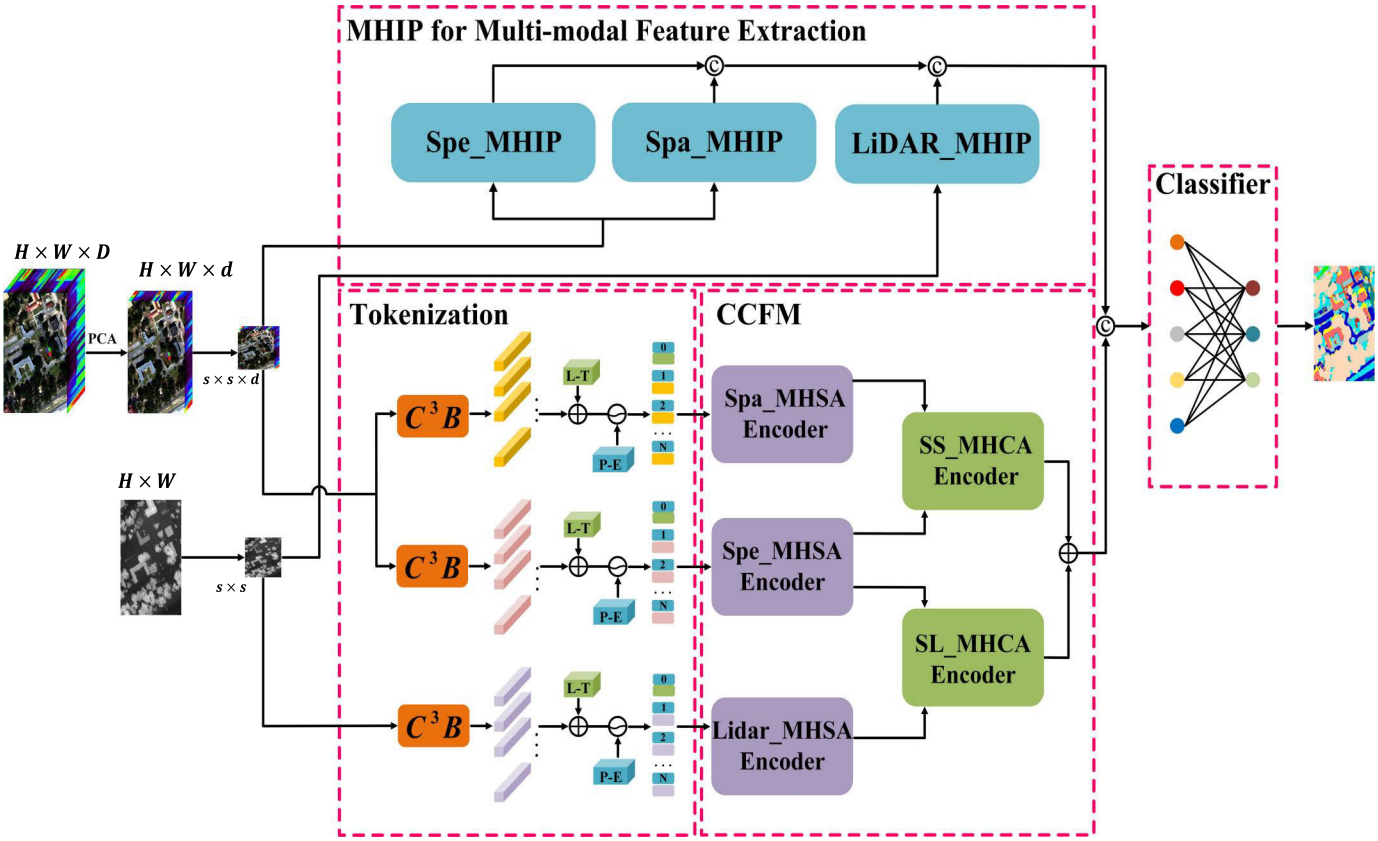


Fig. 2. Overall structure of the MSRCFNet. The proposed MSRCFNet is divided into two branches. In the upper branch, the spatial–spectral features of the HSI and the elevation spatial features of the LiDAR are, respectively, extracted and fused through three MHIP modules. In the lower branch of the proposed network, the HSI and LiDAR data are first tokenized. Subsequently, the tokens of different modalities are input into the CCFM for cross-modal feature fusion. Finally, the features of the upper and lower branches are globally fused and input into the classifier to obtain the final classification result.

with a width of $(s-1)/2$. Assuming there are T labeled training samples, denoted as $\boldsymbol{X} = \{\boldsymbol{X}^i | (\boldsymbol{X}_H^i, \boldsymbol{X}_L^i)\}_{i=1}^T$ with corresponding labels $\Upsilon = \{y^i\}_{i=1}^T$, where $y^i \in \{1, 2, \ldots, U\}$ and U is the number of land-cover types. The remaining samples are separated into training, validation, and testing sets after the zero-labeled pixel blocks have been eliminated.

A. MHIP-Based Multiscale Feature Extraction

In neuroscience, the receptive field refers to a specific region of input data that a neuron is capable of processing or responding to. In other words, a smaller receptive field captures lower level local texture features, while a larger receptive field captures more global contextual information. Based on this, a novel MHIP module is developed to extract multiscale features. As shown in Fig. 3, the MHIP module contains multiple CSRBs based on pseudo-3-D convolutions at different scales. The MHIP module starts to extract global features with relatively large convolution kernels. Subsequently, the extracted global features are integrated downward, and detailed features are extracted through CSRBs with smaller convolution kernels. During the processing of CSRBs at different levels, these features are combined and integrated downward, presenting an inverted pyramid shape that is narrow at the bottom and wide at the top. The CSRBs at various scales effectively extract both local texture features and global context features. Traditional 3-D convolutions handle spatial and spectral dimensions simultaneously, resulting in a high computational cost when processing high-dimensional HSI and LiDAR data. In contrast, pseudo-3-D convolutions separate spectral and spatial feature extraction. This isolation reduces redundant computations, cutting down the number of calculations significantly. Consequently, our model processes data more efficiently with fewer resources. Moreover, the independent extraction allows for a better capture of the unique characteristics of HSI and LiDAR data, enhancing the classification performance. The implementation details of the MHIP module are shown in Table I.

In the Spe_MHIP module, the two CSRBs in the first layer employ convolution kernels of size $k_1^{\text{Spe}} : (1 \times 1 \times 7)$, while the two CSRBs in the second layer employ kernels of size $k_2^{\text{Spe}} : (1 \times 1 \times 5)$, and the two CSRBs in the third layer employ kernels of size $k_3^{\text{Spe}} : (1 \times 1 \times 3)$. The structure of the Spa_MHIP module employed for extracting spatial multiscale features is the same as that of the LiDAR_MHIP module employed for extracting multiscale elevation spatial features from LiDAR data, with the convolution kernel sizes of each CSRB in each layer being $k_1^{\text{Spa}} | k_1^L : (7 \times 7 \times 1)$, $k_2^{\text{Spa}} | k_2^L : (5 \times 5 \times 1)$, and $k_3^{\text{Spa}} | k_3^L : (3 \times 3 \times 1)$, respectively. By adjusting the sizes of the convolution kernels, each layer of the CSRB adapts to the feature extraction requirements at different scales, thereby enhancing the ability of the network to process HSI and LiDAR data. This refined structural configuration makes

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TABLE I
DETAILED STEPS OF THE SPE_MHIP MODULE

Input Size	Module Name	Operations	Output Size
$\boldsymbol{X}_h^C : (1, s \times s \times d)$	Conv-block	Conv3D_BN_Mish	$\boldsymbol{F}_H : (30, s \times s \times d)$
$\boldsymbol{F}_H : (30, s \times s \times d)$	Spe_MHIP	CSRB1_1	$\boldsymbol{F}_{1_1} : (30, s \times s \times d)$
$\boldsymbol{F}_H : (30, s \times s \times d)$	Spe_MHIP	CSRB2_1	$\boldsymbol{F}_{2_1} : (30, s \times s \times d)$
$\boldsymbol{F}_H : (30, s \times s \times d)$	Spe_MHIP	CSRB3_1	$\boldsymbol{F}_{3_1} : (30, s \times s \times d)$
$\boldsymbol{F}_{1_1} : (30, s \times s \times d)$	Spe_MHIP	CSRB1_2	$\boldsymbol{F}_{1_2} : (30, s \times s \times d)$
$\boldsymbol{F}_{1_1} : (30, s \times s \times d)$	Spe_MHIP	Concatenate	$\boldsymbol{F}_1^{fused} : (60, s \times s \times d)$
$\boldsymbol{F}_{2_1} : (30, s \times s \times d)$			
$\boldsymbol{F}_1^{fused} : (60, s \times s \times d)$	Spe_MHIP	CSRB2_2	$\boldsymbol{F}_{2_2} : (60, s \times s \times d)$
$\boldsymbol{F}_1^{fused} : (60, s \times s \times d)$	Spe_MHIP	Concatenate	$\boldsymbol{F}_2^{fused} : (60, s \times s \times d)$
$\boldsymbol{F}_{3_1} : (30, s \times s \times d)$			
$\boldsymbol{F}_2^{fused} : (60, s \times s \times d)$	Spe_MHIP	CSRB3_2	$\boldsymbol{F}_{3_2} : (60, s \times s \times d)$
$\boldsymbol{F}_{1_2} : (30, s \times s \times d)$			
$\boldsymbol{F}_{2_2} : (60, s \times s \times d)$	Spe_MHIP	Concatenate	$\boldsymbol{F}_{out} : (150, s \times s \times d)$
$\boldsymbol{F}_{3_2} : (60, s \times s \times d)$			
$\boldsymbol{F}_{out} : (150, s \times s \times d)$	Spe_MHIP	Reshape	$\boldsymbol{F}_{out}^{spe} : (25, s \times s)$

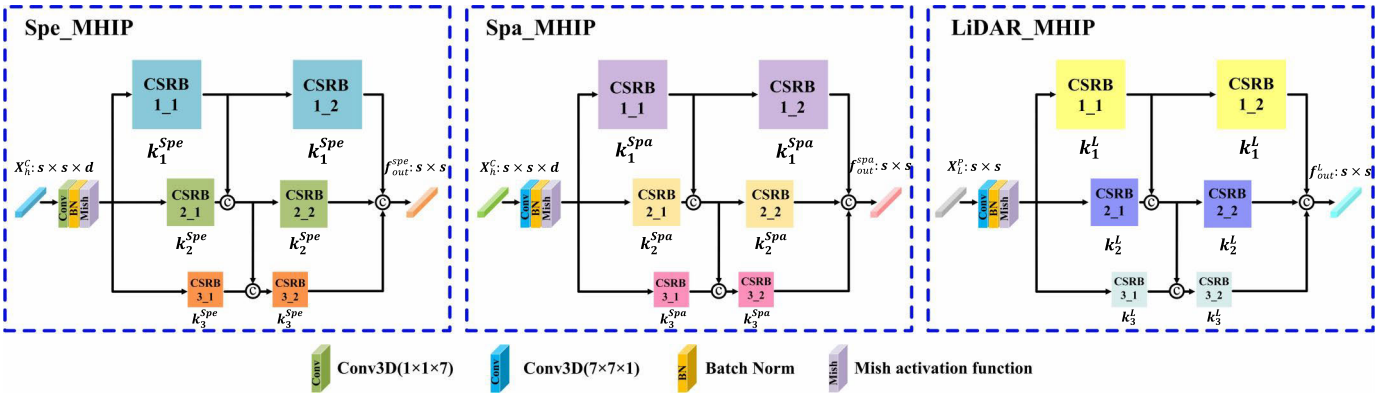


Fig. 3. Structure of Spe_MHIP, Spa_MHIP, and LiDAR_MHIP. The three MHIP modules adopt a hierarchical "inverted pyramid" architectural structure, which contains multiple CSRBs of different scales internally, enabling the full extraction and fusion of multiscale features.

the MHIP module more flexible and effective in handling complex data. The processing procedures of the three MHIP modules are identical; therefore, the implementation details of the MHIP module employed for extracting the spectral features of HSI are specifically described.

The $\boldsymbol{X}_h^C \in \mathbb{R}^{s\times s\times d}$ is first fed into a standard convolutional block of size 30@1 × 1 ×7 to increase the number of its channels, and the output feature is $\boldsymbol{F}_H \in \mathbb{R}^{s\times s\times d}$. Subsequently, $\boldsymbol{F}_H$ is fed into CSRB1_1, CSRB1_2, and CSRB1_3, respectively, obtaining features $\boldsymbol{F}_{1_1} \in \mathbb{R}^{s\times s\times d}$, $\boldsymbol{F}_{2_1} \in \mathbb{R}^{s\times s\times d}$, and $\boldsymbol{F}_{3_1} \in \mathbb{R}^{s\times s\times d}$. Then, $\boldsymbol{F}_{1_1}$ is passed to CSRB1_2 for further feature extraction, with the output feature being $\boldsymbol{F}_{1_2} \in \mathbb{R}^{s\times s\times d}$. In addition, $\boldsymbol{F}_{1_1}$ and $\boldsymbol{F}_{2_1} \in \mathbb{R}^{s\times s\times d}$ need to be concatenated to obtain the feature $\boldsymbol{F}_1^{\text{fused}} \in \mathbb{R}^{s\times s\times d}$. Similar to the above operations, first, $\boldsymbol{F}_1^{\text{fused}}$ is fed into CSRB2_2 to obtain the feature $\boldsymbol{F}_{2_2} \in \mathbb{R}^{s\times s\times d}$. Then, $\boldsymbol{F}_1^{\text{fused}}$ and $\boldsymbol{F}_{3_1}$ are concatenated to obtain the feature $\boldsymbol{F}_2^{\text{fused}} \in \mathbb{R}^{s\times s\times d}$, and $\boldsymbol{F}_2^{\text{fused}}$ is fed into CSRB3_2 to obtain the feature $\boldsymbol{F}_{3_2}$. Finally, $\boldsymbol{F}_{1_2}$, $\boldsymbol{F}_{2_2}$, and $\boldsymbol{F}_{3_2}$ are concatenated to obtain the final output feature $\boldsymbol{F}_{\text{out}}^{\text{spe}} \in \mathbb{R}^{s\times s}$. This design aims to comprehensively extract and fuse multiscale features, thereby improving the complementarity among features of varying scales. The above process is summarized by

$$\begin{aligned} \text{Conv}_{3D}^{1\times1\times7}\left(\boldsymbol{X}_h^C\right) &= \phi\left(\text{BN}\left(\boldsymbol{X}_h^c * \boldsymbol{w}^T + b\right)\right) \\ \boldsymbol{F}_H &= \text{Conv}_{3D}^{1\times1\times7}\left(\boldsymbol{X}_h^C\right) \end{aligned} \tag{1}$$

where $\text{Conv}_{3D}^{1\times1\times7}(\cdot)$ denotes a standard 3-D convolutional block, $\phi(\cdot)$ represents the Mish activation function, $\text{BN}(\cdot)$ denotes the BatchNorm, $*$ indicates the convolution operation, $\boldsymbol{W}^T$ stands for the convolutional kernel weights, and b is the bias term

$$\begin{aligned} &\boldsymbol{F}_{1_1} = \text{CSRB}_{1_1}(\boldsymbol{F}_H), \quad \boldsymbol{F}_{2_1} = \text{CSRB}_{2_1}(\boldsymbol{F}_H) \\ &\boldsymbol{F}_{3_1} = \text{CSRB}_{3_1}(\boldsymbol{F}_H) \end{aligned} \tag{2}$$

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$$\boldsymbol{F}_1^{\text{fused}} = \text{Cat}\left(\boldsymbol{F}_{1_1}, \boldsymbol{F}_{2_1}\right), \quad \boldsymbol{F}_2^{\text{fused}} = \text{Cat}\left(\boldsymbol{F}_1^{\text{fused}}, \boldsymbol{F}_{3_1}\right) \tag{3}$$

$$\boldsymbol{F}_{1_2} = \text{CSRB}_{1_2}\left(\boldsymbol{F}_{1_1}\right), \quad \boldsymbol{F}_{2_2} = \text{CSRB}_{2_2}\left(\boldsymbol{F}_1^{\text{fused}}\right)$$

$$\boldsymbol{F}_{3_2} = \text{CSRB}_{3_2}\left(\boldsymbol{F}_2^{\text{fused}}\right) \tag{4}$$

$$\boldsymbol{F}_{\text{out}}^{\text{spe}} = \text{Cat}\left(\boldsymbol{F}_{1_2}, \boldsymbol{F}_{2_2}, \boldsymbol{F}_{3_2}\right). \tag{5}$$

CSRB is composed of multiple pseudo-3-D convolution blocks. Through multiple splitting and reconstruction operations, the network can capture high-resolution, detailed features while retaining the original features. The details of the CSRB are shown in Fig. 4. After the feature $\boldsymbol{F}_H$ output by the standard 3-D convolution block $\text{Conv}_{3D}^{1\times1\times7}(\cdot)$ is fed into the CSRB, first, $\boldsymbol{F}_H$ is divided into P groups, with each group $\boldsymbol{f}_i \in \mathbb{R}^{k,s\times s\times d}$ having $k = c/P$ channels, where $i \in \{1, 2, 3, \ldots, P\}$. Equation (6) summarizes the process

$$\boldsymbol{f}_i = \text{Split}(\boldsymbol{F}_H), \quad i = 1, 2, 3, \ldots, P \tag{6}$$

where Split$(\cdot)$ represents splitting the feature map along the channel dimension.

After splitting $\boldsymbol{F}_H$ into P groups of feature maps, the feature maps of the i = 1-th group are represented by $\boldsymbol{f}_1 \in \mathbb{R}^{k,s\times s\times d}$, which are passed directly to the next layer without further processing. This is done to safeguard the integrity of the original and crucial information within these feature maps. These directly passed feature maps can effectively complement other features that undergo subsequent processing. This complementary relationship enriches the feature representations available to the model, thereby enhancing its ability to accurately understand and classify data. The feature maps from the remaining groups are first processed through a pseudo-3-D convolution as $\text{Conv}_{p3D}^{j}(\cdot)$, where $j \in \{1, 2, \ldots, P-1\}$. The feature map extracted by $\text{Conv}_{p3D}^{j}(\cdot)$ is denoted as $\boldsymbol{f}_j$. Subsequently, it is divided into two subgroups, denoted as $\boldsymbol{f}_{j_1}$ and $\boldsymbol{f}_{j_2}$. Then, $\boldsymbol{f}_{j_2}$ is concatenated with the subsequent group $\boldsymbol{F}_{i+1}$ and passed into $\text{Conv}_{p3D}^{j+1}(\cdot)$. The final output feature map $\boldsymbol{f}_{\text{out}}$ is created by concatenating $\boldsymbol{f}_1$ and each $\boldsymbol{f}_{j_1}$ along the channel dimension. The above process is summarized by the following equations:

$$\boldsymbol{f}_{j_1}, \boldsymbol{f}_{j_2} = \text{Split}\left(\text{Conv}_{p3D}^{j}\left(\boldsymbol{f}_i\right)\right), \quad i = 2, j = 1 \tag{7}$$

$$\boldsymbol{f}_{j_1}, \boldsymbol{f}_{j_2} = \text{Split}\left(\text{Conv}_{p3D}^{j}\left(\text{Cat}\left(\boldsymbol{f}_{j_2}, \boldsymbol{f}_i\right)\right)\right), \quad i \in \{3, \ldots, G\}, j \in \{i-1, \ldots, P-1\} \tag{8}$$

$$\boldsymbol{f}_{\text{out}} = \text{Cat}_{\text{dim}=1}\left(\boldsymbol{f}_i, \left\{\boldsymbol{f}_{j_1}\right\}_{j=1}^{P-1}\right), \quad i = 1. \tag{9}$$

The core of the CSRB consists of two split and recombination operations. In the first split operation, the feature map $\boldsymbol{F}_H$ is split into multiple subfeature maps $\boldsymbol{F}_i$ along the channel dimension. This process refines the band information to capture more fine-grained feature details. The second splitting operation further refines the feature map into $\boldsymbol{f}_{j_1}$ and $\boldsymbol{f}_{j_2}$, enabling the network to capture higher resolution, detailed features while preserving the original features. The first recombination fuses the feature maps between different groups and enhances the ability of the network to represent diverse features. The second recombination operation integrates all the subfeature maps generated through splitting and recovers the global feature representation by combining detailed information from all subfeature maps. Recombination mainly relies on the concatenate operation, merging feature maps from different branches. Convolution operations extract new features, batch normalization (BN) normalizes for better compatibility, and Mish adds nonlinearity, enhancing discriminability. These operations together improve useful information propagation. Recombining split feature maps fuses group-level

Algorithm 1 Pseudo-Code of the MSRCFNet

Require: $\boldsymbol{X}_H, \boldsymbol{X}_L$: Dimension-reduced HSI and corresponding LiDAR data.
Network Hyper-parameters: Adam optimizer,
Learning Rates: 9×10^{-4} (Trento), 3×10^{-4} (Houston2013), 7×10^{-4} (MUUFL);
Batch Size = 64, Training Sample Rate = 1%;
Ensure: Ensembles of classifiers;
Preprocess dimension-reduced HSI to $\boldsymbol{X}_h^C$;
Preprocess original LiDAR data to $\boldsymbol{X}_L^P$;
Ground-truth label Υ;
for $(\boldsymbol{X}_h^C, \boldsymbol{X}_L^P, \Upsilon)$ in `data_loader` **do**
  Multi-scale Hierarchical Inverted-Pyramid (MHIP) Module:
  Perform **Convolutional Split-Recombination Block (CSRB)** to extract multi-scale features $\boldsymbol{f}_{out}$ from $\boldsymbol{F}_H$ (see Equations (1), (6)-(9));
  Fuse multi-scale features using MHIP to obtain $\boldsymbol{F}_{out}^{spe}$ (see Equations (2)-(5));
  Compute $\boldsymbol{F}_{out}^{spa}$ and $\boldsymbol{F}_{out}^{LiDAR}$:

$$\boldsymbol{f}_{out}^{spa}, \boldsymbol{f}_{out}^{LiDAR} = \text{Spa_MHIP}(\boldsymbol{X}_h^C), \text{LiDAR_MHIP}(\boldsymbol{X}_L^P)$$

  Convolutional Channel Compress Block (C^3B) and Tokenization:
  Extract features:

$$\boldsymbol{F}_{\text{spe}}^{\theta} = \text{C}^3\text{B}_{\text{spe}}(\boldsymbol{X}_H^c) : 7 \times 7 \times 25,$$
$$\boldsymbol{F}_{\text{spa}}^{\theta} = \text{C}^3\text{B}_{\text{spa}}(\boldsymbol{X}_H^c) : 7 \times 7 \times 25,$$
$$\boldsymbol{F}_{\text{LiDAR}}^{\theta} = \text{C}^3\text{B}_{\text{LiDAR}}(\boldsymbol{X}_L^p) : 7 \times 7 \times 25,$$
(see Equations (10)-(13))

  Tokenize features to obtain:

$$\boldsymbol{T}_{\text{Spe}}, \boldsymbol{T}_{\text{Spa}}, \boldsymbol{T}_{\text{LiDAR}} : (1 + \text{feature_tokens_num}) \times \text{d}$$

  (see Equations (14)-(16));
  Cross-modal Cooperative Fusion Module (CCFM):
  Perform CCFM to obtain $\boldsymbol{F}_{out}^{fuse}$ (see Equations (17)-(30));
  Concatenate features:

$$\boldsymbol{F}_{\text{out}} = \text{torch.cat}(\boldsymbol{f}_{\text{out}}^{\text{spe}}, \boldsymbol{f}_{\text{out}}^{\text{spa}}, \boldsymbol{f}_{\text{out}}^{\text{LiDAR}}, \boldsymbol{F}_{out}^{fuse})$$

  Full Connection Classification Module:
  Compute loss:

```
loss = CrossEntropy_Loss(nn.Linear(
         nn.Dropout(
           nn.Mish(
             nn.AdaptiveAvgPool(
               nn.BatchNorm(F_out^fuse))))), Υ)
```

  Update trainable parameters;
end for
return Classification Map;

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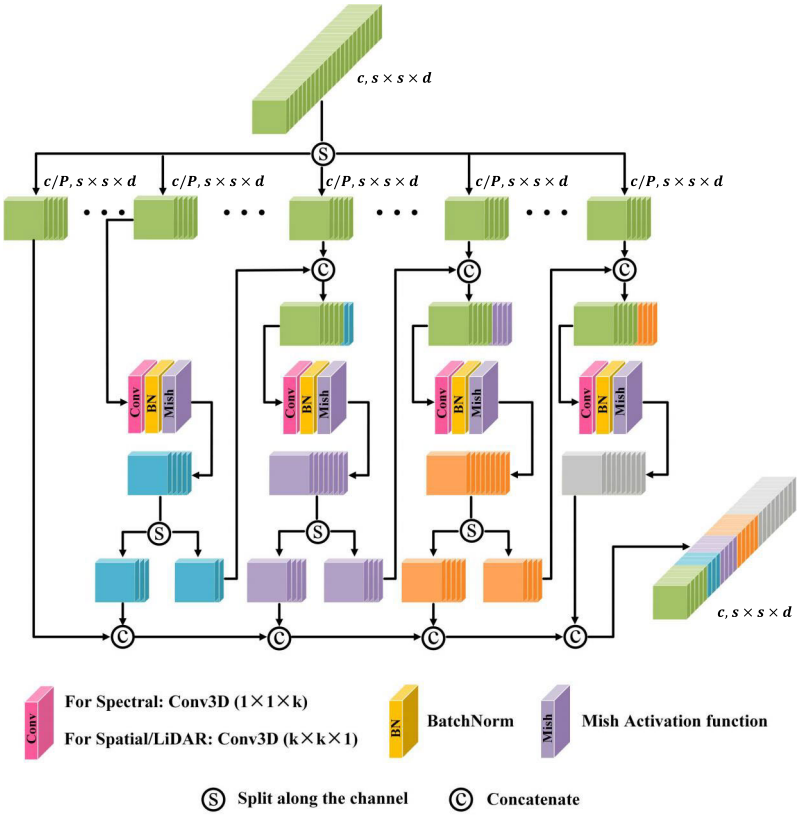


Fig. 4. Structure of CSRB. The CSRB internally consists of multiple pseudo-3-D convolutions. Through a split-and-reconstruct strategy, these convolutions further extract deeper-level features while preserving the original features.

features and integrates subfeature maps, boosting the ability of the network to represent and capture complex features. The implementation details of the CSRB are shown in Table II.

B. Multisource Data Tokenization

HSI contains rich high-dimensional information, with each pixel encompassing numerous spectral bands. High-dimensional data poses challenges for deep learning techniques and increases the complexity of processing. Therefore, we improved the channel attention in reference [62] and designed a convolutional channel compression block (C^3B). During tokenization [63], a large number of channels are typically required for token embedding, and using a fixed-size convolution inevitably leads to significant computational costs. Therefore, a compression factor θ_c is introduced in the two convolutional layers of the compression module to control the extent of channel compression, thereby reducing computational costs.

As illustrated in Fig. 5. Specifically, the C^3B is composed of three 3-D convolutional blocks. First, the first standard 3-D convolutional block $\text{Conv}_{3D}^{1}(\cdot)$ is used to extract shallow-layer features, generating the feature map $\boldsymbol{F}_1$. Subsequently, $\boldsymbol{F}_1$ is successively input into two 3-D convolutional blocks $\text{Conv}_{3D}^{2}(\cdot)$ and $\text{Conv}_{3D}^{3}(\cdot)$ with a compression factor θ_c to generate the feature map $\boldsymbol{F}_2$. Finally, $\boldsymbol{F}_1$ and $\boldsymbol{F}_2$ are element-wise multiplied to enhance the discriminative features and output the feature map $\boldsymbol{F}_3$. This process can be expressed by the following equations:

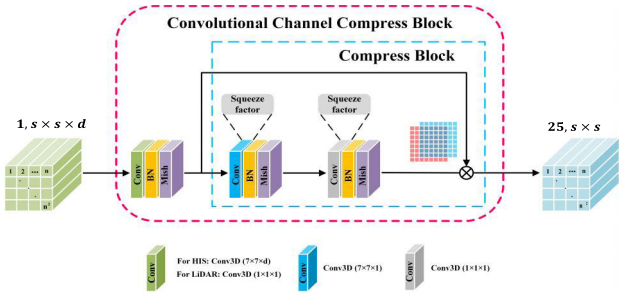


Fig. 5. Structure of C^3B. In the convolution layer of C^3B, a compression factor is introduced to control the compression degree of channels, which effectively enhances the feature expression ability and improves the adaptability of subsequent modules to complex data.

$$\boldsymbol{F}_1 = \text{Conv}_{3D}^{1}(\boldsymbol{X}) \tag{10}$$

$$\boldsymbol{F}_2 = \text{Conv}_{3D}^{3}\left(\text{Conv}_{3D}^{2}(\boldsymbol{F}_1)\right) \tag{11}$$

$$\boldsymbol{F}_3 = \boldsymbol{F}_1 \otimes \boldsymbol{F}_2 \tag{12}$$

$$\text{Conv}_{3D}^{k}(\boldsymbol{X}) = \phi\left(\text{BN}\left(\boldsymbol{X} * \boldsymbol{w}^T * \theta_c + b\right)\right) \tag{13}$$

where ϕ denotes the Mish activation function, $\text{BN}(\cdot)$ denotes the BatchNorm layer, $\boldsymbol{w}^T$ represents the weight of the convolutional kernel, b represents the bias, and θ_c represents the compression factor to control the degree of channel compression.

A tokenization operation is required before embedding the output features of C^3B into the subsequent fusion module.

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TABLE II
DETAILED STEPS OF THE CSRB (TAKING CSRB1_1 AS AN EXAMPLE)

Input Size	Module Name	Operations	Output Size
$\boldsymbol{F}_H : (30, s \times s \times d)$	CSRB1_1	Split_1	$\boldsymbol{f}_i : (6, s \times s \times d), i \in \{1, 2, 3, \ldots, 5\}$
$\boldsymbol{f}_1 : (6, s \times s \times d)$	CSRB1_1	/	$\boldsymbol{f}_1 : (6, s \times s \times d)$
$\boldsymbol{f}_2 : (6, s \times s \times d)$	CSRB1_1	Conv_{p3D}_BN_Mish	$\boldsymbol{f}_2 : (6, s \times s \times d)$
$\boldsymbol{f}_2 : (6, s \times s \times d)$	CSRB1_1	Split_{2_2}	$\boldsymbol{f}_{2_1} : (5, s \times s \times d), \boldsymbol{f}_{2_2} : (4, s \times s \times d)$
$\boldsymbol{f}_3 : (6, s \times s \times d)$	CSRB1_1	Concatenate	$\boldsymbol{f}_1^{fus} : (10, s \times s \times d)$
$\boldsymbol{f}_{2_2} : (4, s \times s \times d)$			
$\boldsymbol{f}_1^{fus} : (10, s \times s \times d)$	CSRB1_1	Conv_{p3D}_BN_Mish	$\boldsymbol{f}_3 : (10, s \times s \times d)$
$\boldsymbol{f}_3 : (10, s \times s \times d)$	CSRB1_1	Split_{3_2}	$\boldsymbol{f}_{3_1} : (5, s \times s \times d), \boldsymbol{f}_{3_2} : (5, s \times s \times d)$
$\boldsymbol{f}_4 : (6, s \times s \times d)$			
$\boldsymbol{f}_{3_2} : (5, s \times s \times d)$	CSRB1_1	Concatenate	$\boldsymbol{f}_2^{fus} : (11, s \times s \times d)$
$\boldsymbol{f}_2^{fus} : (11, s \times s \times d)$	CSRB1_1	Conv_{p3D}_BN_Mish	$\boldsymbol{f}_4 : (11, s \times s \times d)$
$\boldsymbol{f}_4 : (11, s \times s \times d)$	CSRB1_1	Split_{4_2}	$\boldsymbol{f}_{4_1} : (6, s \times s \times d), \boldsymbol{f}_{4_2} : (5, s \times s \times d)$
$\boldsymbol{f}_5 : (6, s \times s \times d)$			
$\boldsymbol{f}_{4_2} : (6, s \times s \times d)$	CSRB1_1	Concatenate	$\boldsymbol{f}_3^{fus} : (12, s \times s \times d)$
$\boldsymbol{f}_3^{fus} : (12, s \times s \times d)$	CSRB1_1	Conv_{p3D}_BN_Mish	$\boldsymbol{f}_5 : (12, s \times s \times d)$
$\boldsymbol{f}_1 : (6, s \times s \times d)$			
$\boldsymbol{f}_{2_1} : (5, s \times s \times d)$			
$\boldsymbol{f}_{3_1} : (5, s \times s \times d)$	CSRB1_1	Concatenate + Down Sampling	$\boldsymbol{f}_{out} : (30, s \times s \times d)$
$\boldsymbol{f}_{4_1} : (6, s \times s \times d)$			
$\boldsymbol{f}_5 : (12, s \times s \times d)$			

As depicted in Fig. 2, the spectral, spatial, and elevation features are each flattened into 1-D vectors. An additional learnable cls-token $\vec{\boldsymbol{t}}_{\text{cls}}$, needs to be appended before each patch token to aggregate all feature information through the attention mechanism. Finally, the sequence information of spectral, spatial, and LiDAR-embedded markers is preserved through a positional embedding $\mathcal{P}(\cdot)$ method that encodes the position of each marker, effectively capturing the positional information in the sequence.

The spectral feature token, denoted by $\boldsymbol{T}_{\text{Spe}} \in \mathbb{R}^{(N+1)\times d}$, the spatial feature token, denoted by $\boldsymbol{T}_{\text{Spa}} \in \mathbb{R}^{(N+1)\times d}$, and the LiDAR feature token, denoted by $\boldsymbol{T}_{\text{LiDAR}} \in \mathbb{R}^{(N+1)\times d}$, are obtained by performing tokenization operations on each 1-D vector separately. The above process is expressed by

$$\boldsymbol{T}_{\text{Spe}} = \mathcal{P}\big(\text{Cat}\big(\vec{\boldsymbol{t}}_{\text{Spe_cls}}, \big(\hat{\boldsymbol{t}}_1, \hat{\boldsymbol{t}}_2, \hat{\boldsymbol{t}}_3, \ldots, \hat{\boldsymbol{t}}_N\big)\big)\big) \tag{14}$$

$$\boldsymbol{T}_{\text{Spa}} = \mathcal{P}\big(\text{Cat}\big(\vec{\boldsymbol{t}}_{\text{Spa_cls}}, \big(\bar{\boldsymbol{t}}_1, \bar{\boldsymbol{t}}_2, \bar{\boldsymbol{t}}_3, \ldots, \bar{\boldsymbol{t}}_N\big)\big)\big) \tag{15}$$

$$\boldsymbol{T}_{\text{LiDAR}} = \mathcal{P}\big(\text{Cat}\big(\vec{\boldsymbol{t}}_{L_\text{cls}}, \big(\tilde{\boldsymbol{t}}_1, \tilde{\boldsymbol{t}}_2, \tilde{\boldsymbol{t}}_3, \ldots, \tilde{\boldsymbol{t}}_N\big)\big)\big). \tag{16}$$

This method effectively preserves feature information during the learning process of the network, enabling the smooth transfer of features to the subsequent fusion module. The implementation details of the C^3B and Tokenization are shown in Table III.

C. Cross-Modal Cooperative Fusion

Due to the presence of substantial complementary information between multimodal data, feature fusion of heterogeneous data effectively leverages the unique advantages of different modalities, thereby enhancing the ability of the network to discriminate ground features. Therefore, a CCFM is designed to integrate HSI and LiDAR data. As shown in Fig. 6, CCFM is divided into two phases. The first phase relies on three encoders of MHSA to model global relationships between the spectral, spatial, and elevation features, respectively. This phase models long-range dependencies within different modal data through the self-attention mechanism to generate richer spectral, spatial, and elevation feature representations, thereby providing global information for the deep feature fusion in the second phase. The implementation details of the CCFM are shown in Table IV.

The three feature embeddings $\boldsymbol{T}_{\text{Spe}}$, $\boldsymbol{T}_{\text{Spa}}$, and $\boldsymbol{T}_{\text{LiDAR}}$ are fed into the encoder consisting of MHSA, LayerNorm (LN), and multilayer perceptron (MLP), respectively. Each branch employs its own set of projection matrices $\boldsymbol{w}_q^x$, $\boldsymbol{w}_k^x$, and $\boldsymbol{w}_v^x$ to preserve modality-specific characteristics. The feature tokens are multiplied by each of these learnable weights and are subsequently transformed into three different weight matrices, referred to as "query" ($\boldsymbol{Q}_x$), "key" ($\boldsymbol{K}_x$), and "value" ($\boldsymbol{V}_x$)

$$\boldsymbol{Q}_{\text{Spe}} = \boldsymbol{T}_{\text{Spe}}\boldsymbol{w}_q^{\text{Spe}}\boldsymbol{K}_{\text{Spe}} = \boldsymbol{T}_{\text{Spe}}\boldsymbol{w}_k^{\text{Spe}}\boldsymbol{V}_{\text{Spe}} = \boldsymbol{T}_{\text{Spe}}\boldsymbol{w}_v^{\text{Spe}} \tag{17}$$

$$\boldsymbol{Q}_{\text{Spa}} = \boldsymbol{T}_{\text{Spa}}\boldsymbol{w}_q^{\text{Spa}}\boldsymbol{K}_{\text{Spa}} = \boldsymbol{T}_{\text{Spa}}\boldsymbol{w}_k^{\text{Spa}}\boldsymbol{V}_{\text{Spa}} = \boldsymbol{T}_{\text{Spa}}\boldsymbol{w}_v^{\text{Spa}} \tag{18}$$

$$\boldsymbol{Q}_L = \boldsymbol{T}_{\text{LiDAR}}\boldsymbol{w}_q^L\boldsymbol{K}_L = \boldsymbol{T}_{\text{LiDAR}}\boldsymbol{w}_k^L\boldsymbol{V}_L = \boldsymbol{T}_{\text{LiDAR}}\boldsymbol{w}_v^L. \tag{19}$$

$\boldsymbol{Q}_x$ and $\boldsymbol{K}_x$ are utilized to compute the attention scores, and softmax transforms these scores into probability weights. The

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TABLE III
DETAILED STEPS OF THE C^{3}B AND TOKENIZATION

Input Size	Module Name	Operations	Output Size
$\boldsymbol{X}_h^C : (1, s \times s \times d)$	C^{3}B	Conv3D_BN_Mish	$\boldsymbol{F}_{Spe} : (25, s \times s)$
$\boldsymbol{X}_h^C : (1, s \times s \times d)$	C^{3}B	Conv3D_BN_Mish	$\boldsymbol{F}_{Spa} : (25, s \times s)$
$\boldsymbol{X}_h^P : (1, s \times s)$	C^{3}B	Conv3D_BN_Mish	$\boldsymbol{F}_{LiDAR} : (25, s \times s)$
$\boldsymbol{F}_{Spe} : (25, s \times s)$	Tokenization	Flatten+cls+Positional	$\boldsymbol{T}_{Spe} : (s \times s + 1, 64)$
$\boldsymbol{F}_{Spa} : (25, s \times s)$	Tokenization	Flatten+cls+Positional	$\boldsymbol{T}_{Spa} : (s \times s + 1, 64)$
$\boldsymbol{F}_{LiDAR} : (25, s \times s)$	Tokenization	Flatten+cls+Positional	$\boldsymbol{T}_{LiDAR} : (s \times s + 1, 64)$

TABLE IV
DETAILED STEPS OF THE CCFM

Input Size	Module Name	Operations	Output Size
$\boldsymbol{T}_{Spe} : (s \times s + 1, 64)$	CCFM	MHSA	$\boldsymbol{T}_{Spe}^2 : (s \times s + 1, 64)$
$\boldsymbol{T}_{Spa} : (s \times s + 1, 64)$	CCFM	MHSA	$\boldsymbol{T}_{Spa}^2 : (s \times s + 1, 64)$
$\boldsymbol{T}_{LiDAR} : (s \times s + 1, 64)$	CCFM	MHSA	$\boldsymbol{T}_{LiDAR}^2 : (s \times s + 1, 64)$
$\boldsymbol{T}_{Spe}^2 : (s \times s + 1, 64)$	CCFM	MHCA	$\boldsymbol{T}_{SS}^2 : (s \times s + 1, 64)$
$\boldsymbol{T}_{Spa}^2 : (s \times s + 1, 64)$	CCFM	MHCA	$\boldsymbol{T}_{SL}^2 : (s \times s + 1, 64)$
$\boldsymbol{T}_{Spe}^2 : (s \times s + 1, 64)$	CCFM	MHCA	$\boldsymbol{T}_{SL}^2 : (s \times s + 1, 64)$
$\boldsymbol{T}_{LiDAR}^2 : (s \times s + 1, 64)$	CCFM	MHCA	$\boldsymbol{T}_{SL}^2 : (s \times s + 1, 64)$
$\boldsymbol{T}_{SS}^2 : (s \times s + 1, 64)$	CCFM	Element-wise Addition	$\boldsymbol{T}_{fused} : (s \times s + 1, 64)$
$\boldsymbol{T}_{SL}^2 : (s \times s + 1, 64)$	CCFM	Element-wise Addition	$\boldsymbol{T}_{fused} : (s \times s + 1, 64)$
$\boldsymbol{T}_{fused} : (s \times s + 1, 64)$	CCFM	Reshape	$\boldsymbol{F}_{out}^{fuse} : (25, s \times s)$

MHSA process is formalized as follows:

$$\mathrm{SA}\big(\boldsymbol{Q}_x, \boldsymbol{K}_x, \boldsymbol{V}_x\big) = \mathrm{Softmax}\left(\frac{\boldsymbol{Q}_x \boldsymbol{K}_x^T}{\sqrt{d_{\boldsymbol{K}_x}}}\right)\boldsymbol{V}_x \tag{20}$$

$$\mathrm{MHSA}\big(\boldsymbol{Q}_x, \boldsymbol{K}_x, \boldsymbol{V}_x\big) = \mathrm{Cat}(\mathrm{SA}_1, \mathrm{SA}_2, \ldots, \mathrm{SA}_h)\boldsymbol{w}^o \tag{21}$$

where SA denotes the self-attention mechanism, $d_{\boldsymbol{K}}$ represents the dimensionality of $\boldsymbol{K}_x$, h is the number of attention heads, and $\boldsymbol{w}^o$ denotes a parameter matrix. Encoder is used to encode the features of the different modal sequences, which is denoted by

$$\begin{aligned}\boldsymbol{T}_{\mathrm{Spa}}^1 &= \mathrm{MHSA}\big(\mathrm{LN}\big(\boldsymbol{T}_{\mathrm{Spa}}\big)\big) + \boldsymbol{T}_{\mathrm{Spa}} \\ \boldsymbol{T}_{\mathrm{Spa}}^2 &= \mathrm{MLP}\Big(\mathrm{LN}\Big(\boldsymbol{T}_{\mathrm{Spa}}^1\Big)\Big) + \boldsymbol{T}_{\mathrm{Spa}}^1\end{aligned} \tag{22}$$

$$\begin{aligned}\boldsymbol{T}_{\mathrm{Spe}}^1 &= \mathrm{MHSA}\big(\mathrm{LN}\big(\boldsymbol{T}_{\mathrm{Spe}}\big)\big) + \boldsymbol{T}_{\mathrm{Spe}} \\ \boldsymbol{T}_{\mathrm{Spe}}^2 &= \mathrm{MLP}\Big(\mathrm{LN}\Big(\boldsymbol{T}_{\mathrm{Spe}}^1\Big)\Big) + \boldsymbol{T}_{\mathrm{Spe}}^1\end{aligned} \tag{23}$$

$$\begin{aligned}\boldsymbol{T}_{\mathrm{LiDAR}}^1 &= \mathrm{MHSA}(\mathrm{LN}(\boldsymbol{T}_{\mathrm{LiDAR}})) + \boldsymbol{T}_{\mathrm{LiDAR}} \\ \boldsymbol{T}_{\mathrm{LiDAR}}^2 &= \mathrm{MLP}\big(\mathrm{LN}\big(\boldsymbol{T}_{\mathrm{LiDAR}}^1\big)\big) + \boldsymbol{T}_{\mathrm{LiDAR}}^1\end{aligned} \tag{24}$$

specifically, $\boldsymbol{T}_{\mathrm{Spa}}^2$, $\boldsymbol{T}_{\mathrm{Spe}}^2$, and $\boldsymbol{T}_{\mathrm{LiDAR}}^2$ are, respectively, the outputs of the MHSA encoder.

In the first phase, three MHSA-based transformer encoders are designed to capture the long-range dependency relationships inherent in heterogeneous data, thereby enhancing the ability of the module to distinguish between different features. Since HSI and LiDAR data are captured from the same area at the same spatial resolution, fusing these two types of data can take full advantage of their complementary properties and thus improve classification performance. In the second phase, two multihead cross-attention (MHCA)-based transformer encoders are employed to perform deep fusion of inherent features from two types of data. Specifically, to learn the relationships among feature tokens across different modalities, the two outputs $\boldsymbol{T}_{\mathrm{Spa}}^2$ and $\boldsymbol{T}_{\mathrm{Spe}}^2$ from the first stage are fed into one of the MHCA encoders. Three weight matrices, $\boldsymbol{Q}_{\mathrm{Spa}}$, $\boldsymbol{K}_{\mathrm{Spe}}$, and $\boldsymbol{V}_{\mathrm{Spe}}$, are generated before feeding the feature tokens into the MHCA encoder. These generated weight matrices are then passed through the MHCA for cross-modal feature fusion. The process is summarized by the following equations:

$$\mathrm{CA}\big(\boldsymbol{Q}_{\mathrm{Spa}}, \boldsymbol{K}_{\mathrm{Spe}}, \boldsymbol{V}_{\mathrm{Spe}}\big) = \mathrm{Softmax}\left(\frac{\boldsymbol{Q}_{\mathrm{Spa}} \boldsymbol{K}_{\mathrm{Spe}}^T}{\sqrt{d_{\boldsymbol{K}_{\mathrm{Spe}}}}}\right)\boldsymbol{V}_{\mathrm{Spe}} \tag{25}$$

$$\mathrm{MHCA}\big(\boldsymbol{Q}_{\mathrm{Spa}}, \boldsymbol{K}_{\mathrm{Spe}}, \boldsymbol{V}_{\mathrm{Spe}}\big) = \mathrm{Cat}(\mathrm{CA}_1, \ldots, \mathrm{CA}_h)\boldsymbol{w}^z \tag{26}$$

$$\begin{aligned}\boldsymbol{T}_{\mathrm{SS}}^1 &= \mathrm{MHCA}\Big(\mathrm{LN}\Big(\boldsymbol{T}_{\mathrm{Spa}}^2\Big), \mathrm{LN}\Big(\boldsymbol{T}_{\mathrm{Spe}}^2\Big)\Big) + \boldsymbol{T}_{\mathrm{Spa}}^2 + \boldsymbol{T}_{\mathrm{Spe}}^2 \\ \boldsymbol{T}_{\mathrm{SS}}^2 &= \mathrm{MLP}\big(\mathrm{LN}\big(\boldsymbol{T}_{\mathrm{SS}}^1\big)\big) + \boldsymbol{T}_{\mathrm{SS}}^1.\end{aligned} \tag{27}$$

Similarly, the outputs $\boldsymbol{T}_{\mathrm{Spe}}^2$ and $\boldsymbol{T}_{\mathrm{LiDAR}}^2$ are fed into another MHCA encoder. Three weight matrices $\boldsymbol{Q}_{\mathrm{Spe}}$, $\boldsymbol{K}_{\mathrm{LiDAR}}$, and $\boldsymbol{V}_L$ are generated and used for cross-modal feature fusion

$$\mathrm{CA}\big(\boldsymbol{Q}_{\mathrm{Spe}}, \boldsymbol{K}_L, \boldsymbol{V}_L\big) = \mathrm{Softmax}\left(\frac{\boldsymbol{Q}_{\mathrm{Spe}} \boldsymbol{K}_L^T}{\sqrt{d_{\boldsymbol{K}_L}}}\right)\boldsymbol{V}_L \tag{28}$$

$$\mathrm{MHCA}\big(\boldsymbol{Q}_{\mathrm{Spe}}, \boldsymbol{K}_L, \boldsymbol{V}_L\big) = \mathrm{Cat}(\mathrm{CA}_1, \ldots, \mathrm{CA}_h)\boldsymbol{w}^u \tag{29}$$

$$\begin{aligned}\boldsymbol{T}_{\mathrm{SL}}^1 &= \mathrm{MHCA}\Big(\mathrm{LN}\Big(\boldsymbol{T}_{\mathrm{Spe}}^2\Big), \mathrm{LN}\big(\boldsymbol{T}_{\mathrm{LiDAR}}^2\big)\Big) + \boldsymbol{T}_{\mathrm{Spe}}^2 + \boldsymbol{T}_{\mathrm{LiDAR}}^2 \\ \boldsymbol{T}_{\mathrm{SL}}^2 &= \mathrm{MLP}\big(\mathrm{LN}\big(\boldsymbol{T}_{\mathrm{SL}}^1\big)\big) + \boldsymbol{T}_{\mathrm{SL}}^1.\end{aligned} \tag{30}$$

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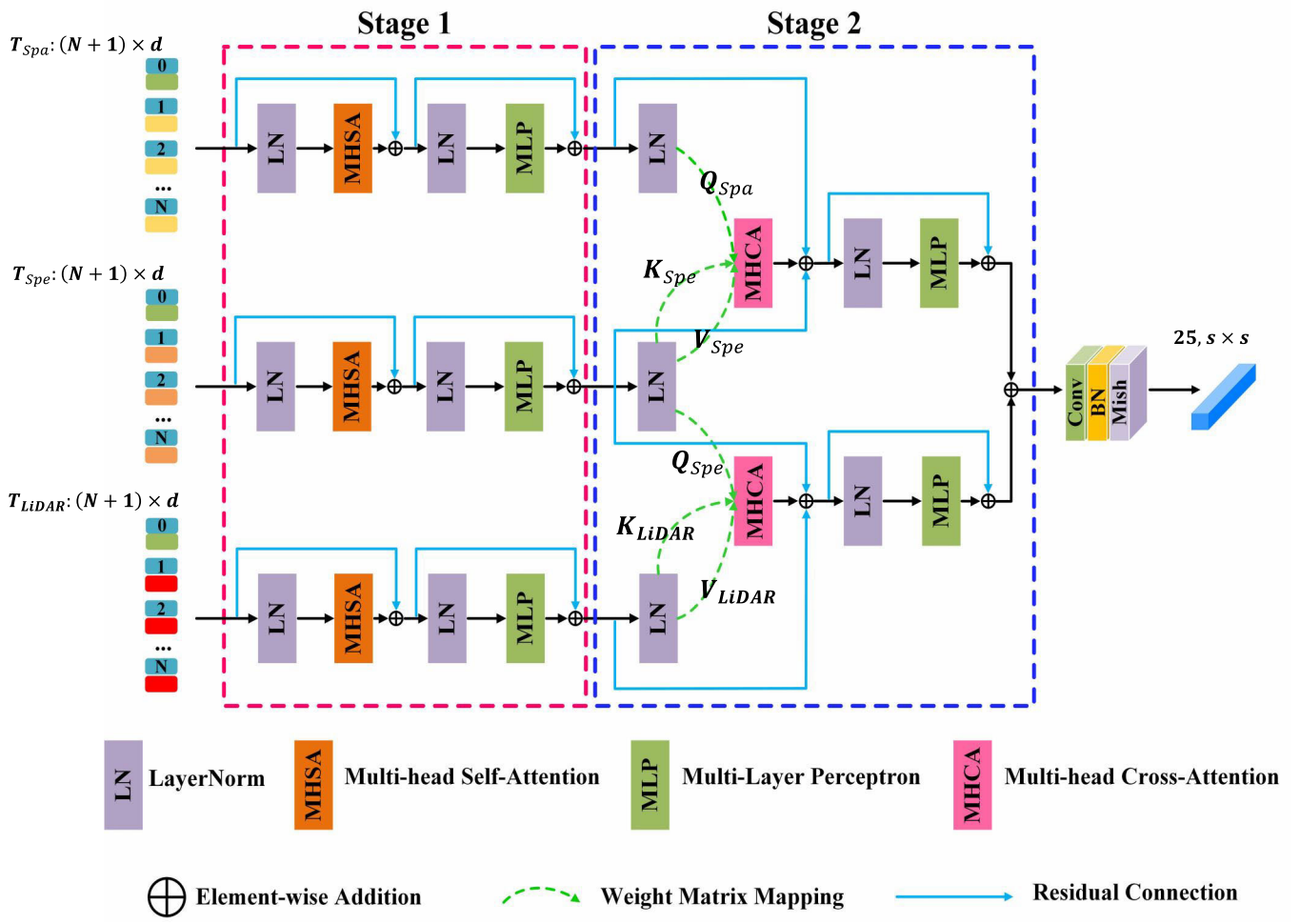


Fig. 6. Structure of CCFM. The CCFM is divided into two stages. In the first stage, self-attention is employed to capture the long-range dependencies within specific modalities. In the second stage, cross-attention is adopted to conduct cross-modal deep feature fusion on heterogeneous features.

Finally, the outputs of the two MHCA-encoders are fused through the element-wise addition $\oplus$. The fused data is fed into a standard 3-D-convolution block $\text{Conv}_{3D}^{s\times s\times d}(\cdot)$ with a kernel size of $25@s \times s \times d$ after a reshape operation $\text{Re}(\cdot)$. This 3-D-convolution block consists of a convolutional layer, a BN layer, and a Mish activation function. The 3-D-convolution block can effectively enhance the representational ability of the model. The operation can be formulated as follows:

$$\boldsymbol{F}_{\text{out}}^{\text{fuse}} = \text{Conv}_{3D}^{s\times s\times d}\left(\text{Re}\left(\boldsymbol{T}_{\text{SS}}^{2} \oplus \boldsymbol{T}_{\text{SL}}^{2}\right)\right). \tag{31}$$

D. Classifier

The features processed by the MHIP module and CCFM, are concatenated and then passed through an adaptive average pooling layer with BN and Mish activation functions. Finally, a linear layer produces the final classification result. Notably, the use of cross-entropy loss to train the network inherently incorporates a probability distribution over the labels, thus eliminating the need for Softmax to produce the final classification results.

IV. Experiment and Analysis

A. Dataset Description

The Trento dataset [64], the Houston2013 dataset [65], and the MUUFL dataset [66], [67] are used to conduct extensive experiments to assess the effectiveness of the proposed MSRCFNet. Detailed descriptions of these three datasets are provided below.

1) *Trento Dataset:* This dataset includes LiDAR DSM data captured by the Optech ALTM 3100 EA sensor and HSI data from the AISA Eagle sensor. The data has a spatial resolution of 1.0 m and dimensions of 166 × 600 pixels. While the LiDAR data provides highly accurate topographic and elevation information, the HSI data consists of 63 bands with a spectral range of 400–990 nm. The ground objects are categorized into six classes. Specific details are provided in Table V and Fig. 7.
2) *Houston2013 Dataset:* The Houston2013 dataset was recorded using the ITRES CASI-1500 sensor in Houston, TX, USA, on the University of Houston campus and the surrounding rural area. The data, primarily

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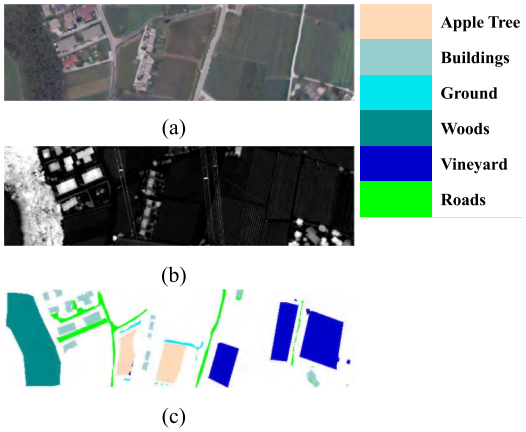


Fig. 7. Trento: (a) pseudo-color image (31, 14, and 2 bands); (b) LiDAR-derived DSM image; and (c) ground-truth map.

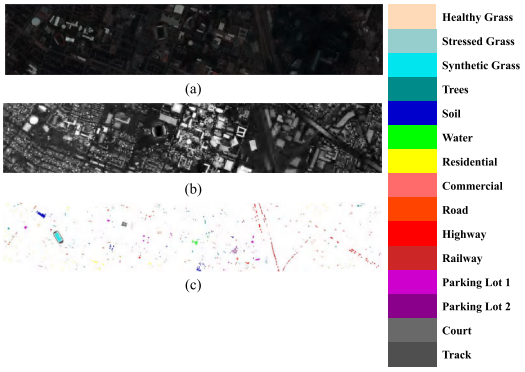


Fig. 8. Houston2013 dataset: (a) pseudo-color image (64, 30, and 20 bands); (b) LiDAR-derived DSM image; and (c) ground-truth map.

from two sources, was made publicly available through IEEE GRSS DFC 2013. With a spatial resolution of 2.5 m and dimensions of 349 × 1905 pixels, the HSI and LiDAR data cover 15 categories. The LiDAR data consists of a single band, while the HSI data includes 144 bands with spectral resolutions ranging from 0.38 to 1.05 μm. Specific details are provided in Table V and Fig. 8.

3) *MUUFL Dataset:* In November 2010, the ITRES CASI-1500 sensor at the University of Southern Mississippi Gulf Park campus in Long Beach, Mississippi, gathered the MUUFL dataset. The dataset has a spatial resolution of 0.54×1.0 m, with spatial dimensions of 325×220 pixels. The HSI consists of 64 spectral bands, covering a range from 375 to 1050 nm, and includes 11 land cover types. Specific details are provided in Table V and Fig. 9.

B. Experiment Setup, Evaluation Metrics and Comparison Algorithms

1) *Configurations:* Experiments on the proposed method and other deep learning approaches are conducted in an environment configured with CUDA v11.6, PyTorch 1.13, and Python 3.8. The tests are executed on laptops featuring an Intel[1] Core[2] i5-1035G1 CPU @ 1.00 GHz, 16-GB RAM, and an NVIDIA GeForce RTX 2080 Ti GPU with 11-GB VRAM. The Adam optimizer is used to optimize the network.

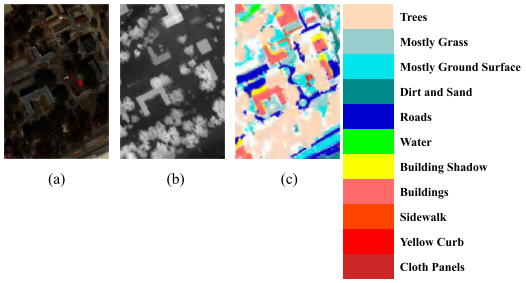


Fig. 9. MUUFL dataset: (a) pseudo-color image (31, 16, and 6 bands); (b) LiDAR-derived DSM image; and (c) ground-truth map.

2) *Evaluation Metrics:* Overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are used as evaluation indicators in the experiments. Specifically, OA is calculated as the ratio of correctly classified samples to the total number of samples and is defined as follows:

$$\mathrm{OA} = \frac{\sum_i C_{ii}}{G}. \tag{32}$$

With equal weight assigned to each category, AA represents the AA across all classes. It is defined as follows:

$$\mathrm{AA} = \frac{\sum_i \mathrm{EA}_i}{N} = \frac{1}{N} \times \frac{\sum_i C_{ii}}{C_{ij}} \tag{33}$$

where EA represents the proportion of samples correctly classified within a specific category. The following defines Kappa, a statistical metric that evaluates the degree of agreement between classification results and ground truth data:

$$\kappa = \frac{\mathrm{OA} - C_c}{1 - C_c} \tag{34}$$

$$C_c = \frac{\sum_n \left(\sum_j C_{nj} \cdot \sum_i C_{in} \right)}{G \cdot G}. \tag{35}$$

3) *Comparison Algorithm:* To comprehensively assess the effectiveness of MSRCFNet, we select eight advanced methods as contrast objects. These methods include HDDA [68], MCFN [29], CoupledCNN [69], S^{2}ENet [70], HCT [40], Sal^{2}RNet [71], MS2CANet [32], and Cross-HL [37]. Among them, HDDA and MCFN are both two-branch HSI classification methods based on CNN and attention mechanisms. CoupledCNN is a classification method for multisource fusion using CNN. S^{2}ENet is an approach to cross-modal information interaction enhancement algorithm based on CNN architecture. HCT is a multisource fusion classification method that integrates CNN and transformer. Sal^{2}RNet is a multisource fusion classification algorithm based on cross-level interactive

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calibration based on CNN. MS2CANet is a multisource method for multisource feature enhancement and calibration based on multiscale CNN. Cross-HL is a multisource cross-attention classification algorithm based on ViT.

C. Parameter Analysis

To assess the optimal performance of the proposed network, the control variable method is used to test the parameters influencing classification results, such as the reduction of spectral dimension, learning rate, input data patch size, and depth of the encoder. The impact of these parameters on network performance is detailed in this section.

1) *Reduced Spectral Dimension:* The dimensionality reduction of the dataset through PCA effectively removes redundant spectral information, which significantly impacts the entire training process and classification outcomes. To assess the effectiveness of PCA, different spectral dimensions are selected from the candidate set {10, 20, 30, 40, 50, 60} for the experiments. Fig. 10 illustrates the OAs, AAs, and Kappa values achieved by the proposed network on three datasets following dimensionality reduction using PCA. The experimental results reveal that the proposed approach achieves optimal performance across the three datasets when the spectral dimensions are reduced to 30.
2) *Learning Rate:* In deep learning, the learning rate controls the step size for optimization. A rate that's too high may hinder convergence, while one that's too low may lead to local optima. Thus, the learning rate determines the range of updated weights in each iteration and is a crucial parameter. Several learning rates are selected from the candidate set {3E-4, 5E-4, 7E-4, 9E-4} to evaluate their impact on network performance. Fig. 11 illustrates the OAs, AAs, and Kappa values achieved by the proposed network on different datasets using various learning rates. The experimental findings demonstrate that when the learning rates are set to 9E-4, 3E-4, and 7E-4, respectively, the suggested approach performs best on the Trento, Houston2013, and MUUFL datasets.
3) *Patch Size:* Smaller patches capture finer local features but may introduce information redundancy, increasing computational cost, while larger patches better capture global features yet often miss finer details. Therefore, in remote sensing image classification, patches of appropriate size are crucial for feature extraction. To assess the effect of patch size, different patch sizes are selected from the candidate set {5 × 5, 7 × 7, 9 × 9, 11 × 11, 13 × 13} for the experiments. Fig. 12 illustrates the impact on OAs, AAs, and Kappa under different patch sizes across the three datasets. When the patch size increases from 5 × 5 to 7 × 7, OA gradually increases. When the patch size is 7 × 7, the OAs of the three datasets are maximized. With the continuous increase of patch size, OAs of different datasets exhibit a decreasing trend. This occurs because large patch sizes may include irrelevant pixels, preventing the capture of fine terrain details and reducing classification performance.
4) *Depth of Encoder:* With the increase of encoder depth, the network can effectively capture higher semantic information in the data, thus extracting more discriminating information. The effect of different encoder depths, selected from the set {1, 3, 6, 8, 12}, on the performance of the proposed network is evaluated. Fig. 13 illustrates the impact of the proposed method using encoders of different depths on the OAs, AAs, and Kappa across the three datasets. When the encoder depth is 6, the OA is highest on the Trento and Houston2013 datasets. When the encoder depth is 8, the highest OA is obtained on the MUUFL dataset. This is because the MUUFL dataset has more training samples, so increasing the encoder depth is necessary to allow the network to achieve optimal performance on the MUUFL dataset.

D. Experimental Results and Analysis

In order to demonstrate the performance of the proposed network, eight comparison methods and MSRCFNet are tested on three datasets, respectively. The training, validation, and testing sets each use distinct data, with training samples selected randomly. The results of each method are averaged over 5 runs. Tables VI–VIII and Figs. 14–16 present the classification results for the three datasets, with the best classification outcomes highlighted in bold in Tables VI–VIII.

1) *Experiment Result Analysis of Trento Dataset:* The Trento dataset has sufficient training samples, and fewer ground object coverage categories, which allows all algorithms to achieve high classification accuracy. Table VI and Fig. 14 present the detailed classification results and the full-pixel classification map, respectively. It can be seen from Table VI, using HSI alone as input for HDDA and MCFN achieves OAs of 96.39% and 97.86%, respectively. In contrast, most multisource fusion methods achieve OAs above 98%. The OA of the proposed MSRCFNet reaches 99.21%, which is the highest among all methods. This suggests that the inclusion of LiDAR data effectively compensates for the limitations of HSI, allowing the network to better capture intricate land cover features and reduce misclassifications. However, compared to HDDA and MCFN, fusion classification algorithms such as CoupledCNN, HCT, Sal^{2}RNet, MS2CANet, and Cross-HL exhibit a decrease in accuracy for C1 and C3 after the introduction of LiDAR elevation information. This may be due to the fact that the integration of LiDAR data affects the classification decision of the central pixel in these methods. However, the proposed MSRCFNet demonstrates good classification performance for both these categories. This suggests that the proposed MSRCFNet is effective in efficiently capturing and fusing the complementary and relevant data from both sources. By observing Fig. 14, we find that the full-pixel classification map produced by the proposed MSRCFNet [see Fig. 14(i)] provides a more accurate representation of land cover details, further demonstrating the effectiveness of the method.

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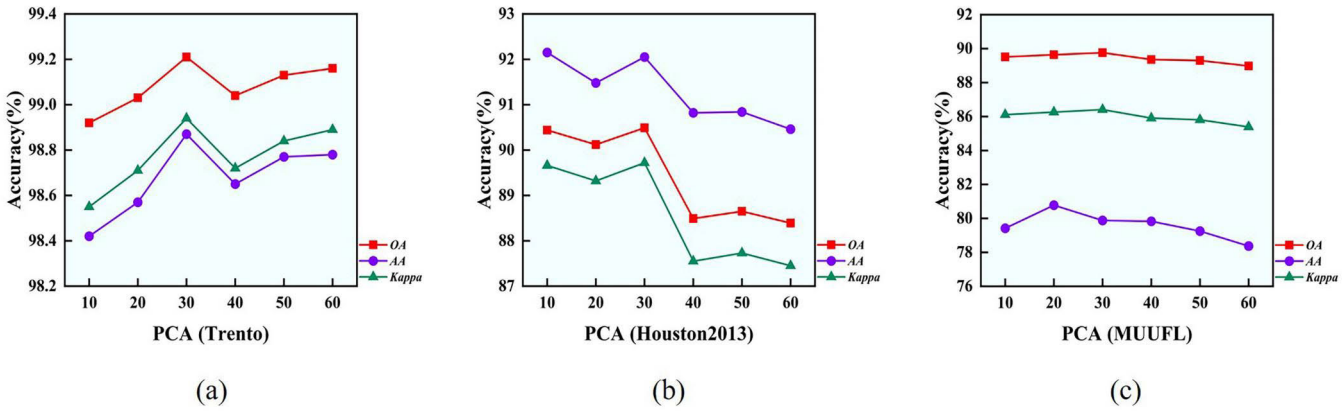


Fig. 10. Impact of different PCA in terms of OA (%), AA (%), and Kappa × 100 (%) on three datasets. (a) Trento dataset. (b) Houston2013 dataset. (c) MUUFL dataset.

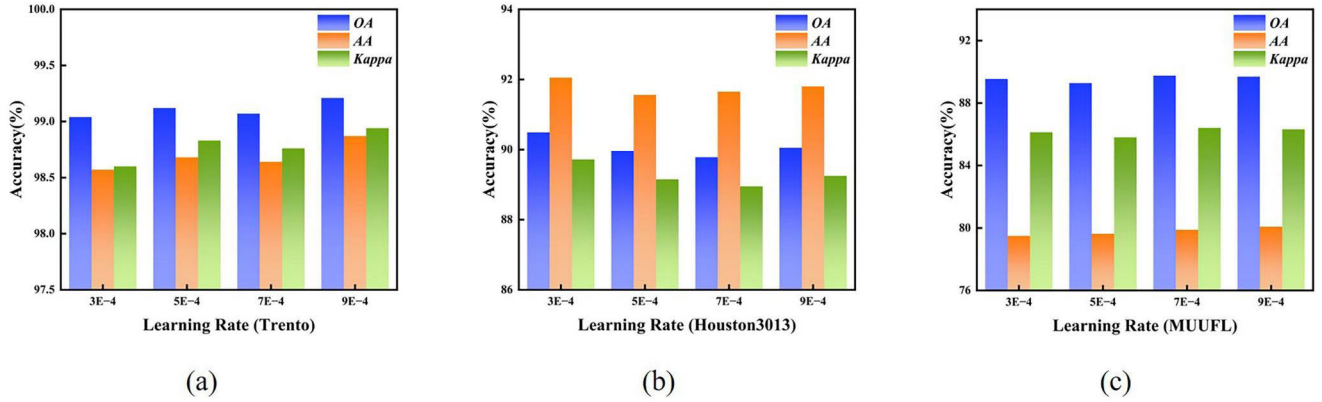


Fig. 11. Impact of different learning rates in terms of OA (%), AA (%), and Kappa × 100 (%) on three datasets. (a) Trento dataset. (b) Houston2013 dataset. (c) MUUFL dataset.

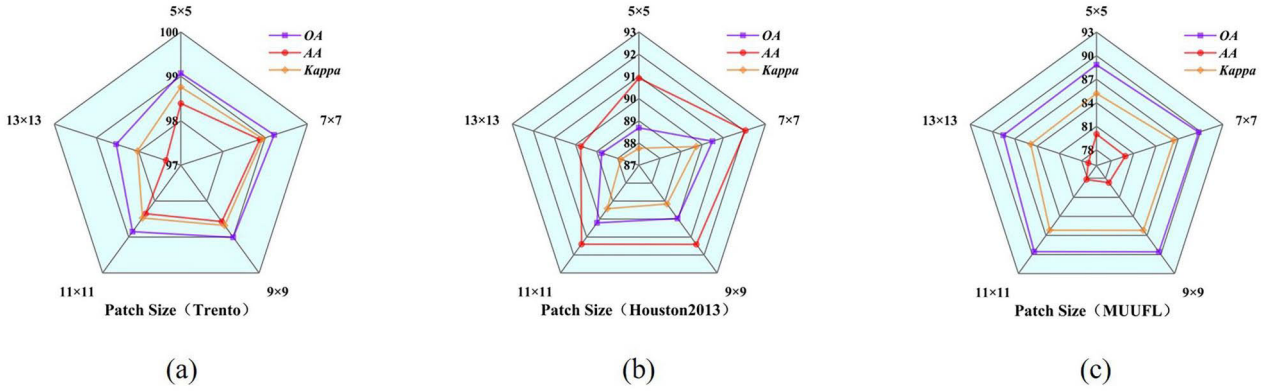


Fig. 12. Impact of different patch sizes in terms of OA (%), AA (%), and Kappa × 100 (%) on three datasets. (a) Trento dataset. (b) Houston2013 dataset. (c) MUUFL dataset.

2) *Experiment Result Analysis of Houston2013 Dataset:* Experiments were performed on this dataset using 1% training samples. The classification results and full-pixel classification maps are presented in Table VII and Fig. 15, respectively. Due to the small sample size and the relatively large number of categories in the Houston2013 dataset, the classification accuracy of most algorithms is relatively low. Nevertheless, the proposed MSRCFNet still achieves satisfactory classification results, with OA, AA, and Kappa reaching 90.49%, 92.05%, and 89.72%, respectively. The OAs of HDDA and MCFN using HSI as input are 87.12% and 87.37%, respectively. Compared to these two methods, the OA of the proposed MSRCFNet improves by 3.37% and 3.12%, respectively. This indicates that the proposed MSRCFNet effectively fuses HSI and LiDAR data, enhancing the ability of the network to recognize detailed features and complex terrain. S^2ENet and HCT

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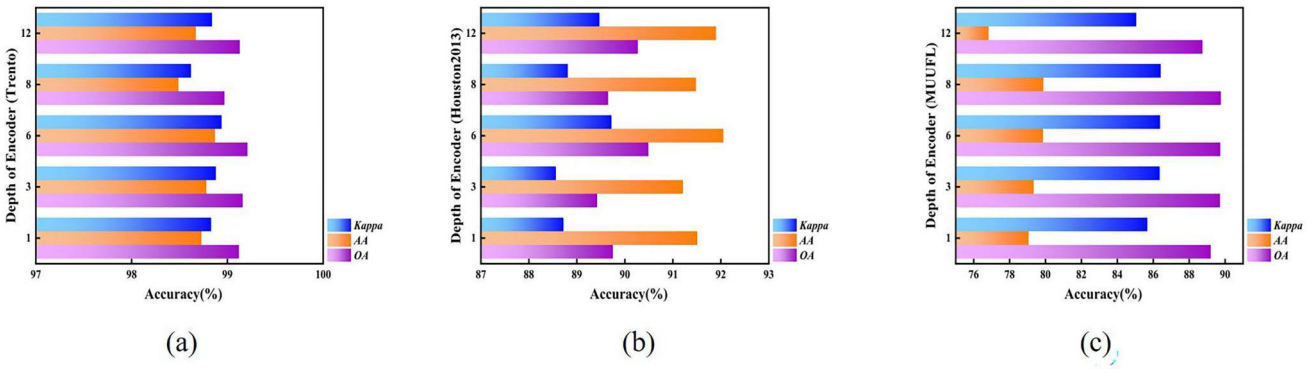


Fig. 13. Impact of different encoder depths in terms of OA (%), AA (%), and Kappa × 100 (%) on three datasets. (a) Trento dataset. (b) Houston2013 dataset. (c) MUUFL dataset.

TABLE V
CLASSES AND THE NUMBER OF SAMPLES IN THE TRENTO, HOUSTON2013, AND MUUFL DATASETS

	Trento dataset				Houston2013 dataset				MUUFL dataset			
No.	Class Name	Train	Val	Test	Class Name	Train	Val	Test	Class Name	Train	Val	Test
C01	**Apple Trees**	41	41	3952	**Healthy Grass**	13	13	1225	**Trees**	233	233	22780
C02	**Buildings**	30	30	2843	**Stressed Grass**	13	13	1228	**Mostly Grass**	43	43	4184
C03	**Ground**	5	5	469	**Synthetic Grass**	7	7	683	**Mixed Ground Surface**	69	69	6744
C04	**Woods**	92	92	8939	**Trees**	13	13	1218	**Dirt and Sand**	19	19	1768
C05	**Vineyard**	106	106	10289	**Soil**	13	13	1216	**Road**	67	67	6753
C06	**Roads**	32	32	3110	**Water**	4	4	317	**Water**	5	5	456
C07					**Residential**	13	13	1242	**Buildings Shadow**	23	23	2187
C08					**Commercial**	13	13	1218	**Buildings**	63	63	6114
C09					**Road**	13	13	1226	**Sidewalk**	14	14	1357
C10					**Highway**	13	13	1201	**Yellow Curb**	2	2	179
C11					**Railway**	13	13	1209	**Cloth Panels**	233	3	263
C12					**Parking Lot1**	13	13	1207				
C13					**Parking Lot2**	5	5	459				
C14					**Tennis Court**	5	5	418				
C15					**Running Track**	7	7	646				
	Total	306	306	29602	**Total**	158	158	14713	**Total**	541	541	52785

TABLE VI
CLASSIFICATION RESULT INCLUDING OA (%), AA (%), AND KAPPA WITH STANDARD DEVIATION FOR THE TRENTO DATASET (OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD)

	HSI Input		HSI and LiDAR Input						
NO.	**HDDA**	**MCFN**	**CoupledCNN**	**S^{2}ENet**	**HCT**	**Sal^{2}RNet**	**MS2CANet**	**Cross-HL**	**Proposed**
C01	99.40±0.39	99.37±0.40	96.21±2.37	99.45±0.29	97.32±1.42	97.48±1.80	97.47±1.96	92.62±2.22	**99.61±0.24**
C02	86.11±5.33	91.45±1.11	95.31±0.69	92.19±2.46	95.92±0.62	96.20±1.22	95.99±0.58	93.57±0.78	**97.15±0.84**
C03	**100.00±0.00**	**100.00±0.00**	99.81±0.27	99.91±0.17	98.35±2.96	96.88±3.46	98.49±2.34	84.77±2.69	**100.00±0.00**
C04	99.80±0.08	99.91±0.02	**100.00±0.00**	99.94±0.04	99.84±0.07	99.87±0.08	**100.00±0.00**	99.32±0.56	**100.00±0.00**
C05	98.63±0.32	99.64±0.03	99.51±0.21	**99.66±0.03**	99.25±0.34	99.05±0.27	99.64±0.06	97.22±0.58	99.61±0.07
C06	90.57±3.49	89.83±2.28	97.17±0.71	93.82±0.80	95.08±0.86	92.75±1.92	97.27±0.69	**97.96±0.94**	96.84±1.01
OA (%)	96.93±0.45	97.86±0.19	98.54±0.33	98.38±0.34	98.38±0.20	98.09±0.40	98.83±0.34	96.85±0.37	**99.21±0.12**
AA (%)	95.75±0.45	96.70±0.25	98.00±0.40	97.50±0.53	97.63±0.48	97.04±0.84	98.14±0.42	94.24±4.11	**98.87±0.18**
Kappa×100	95.91±0.60	97.14±0.25	98.05±0.45	97.84±0.45	97.84±0.27	97.46±0.53	98.44±0.45	95.78±0.51	**98.94±0.16**

achieve good classification results in the fusion classification network, with OA reaching 89.15% and 89.13%, respectively. This may be because S^{2}ENet and HCT effectively combine the key information from both HSI

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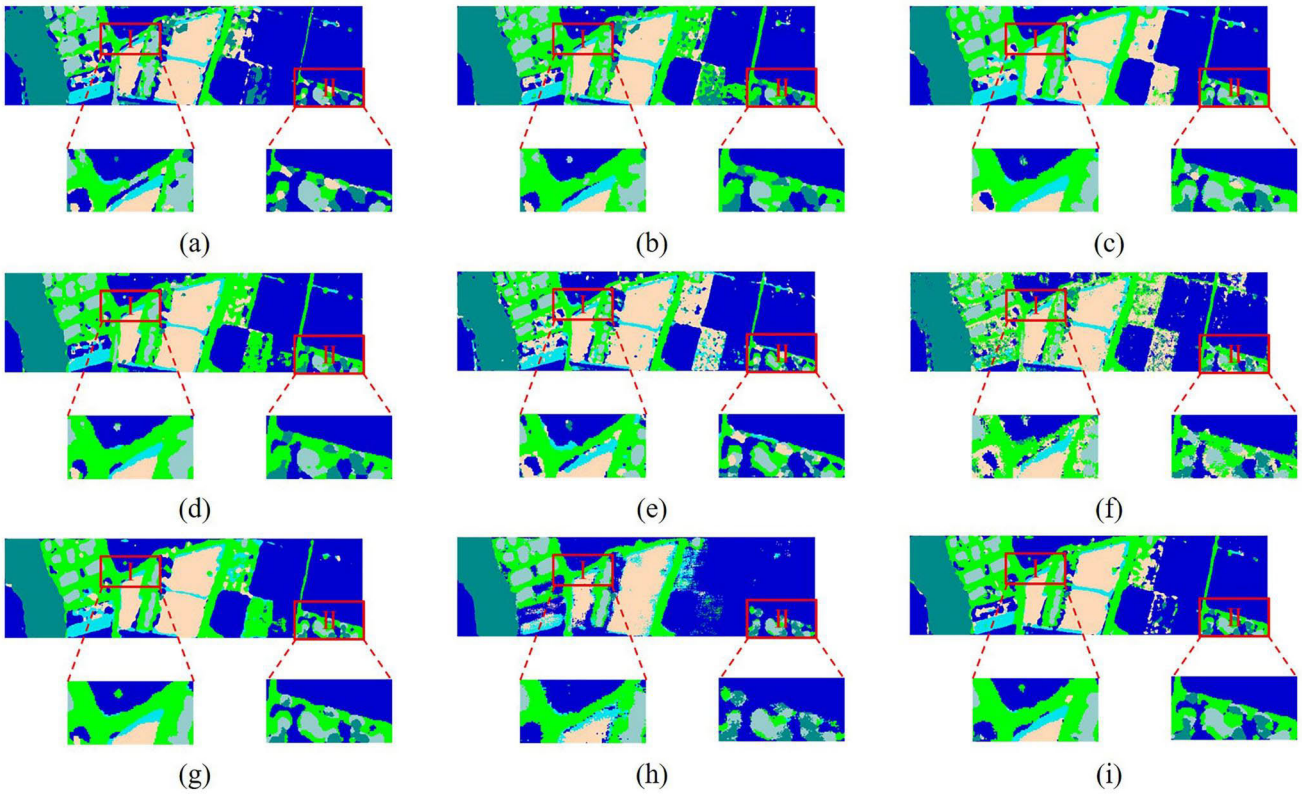


Fig. 14. Full-pixel classification maps for the Trento dataset. (a) HDDA. (b) MCFN. (c) CoupledCNN. (d) S^{2}ENet. (e) HCT. (f) Sal^{2}RNet. (g) MS2CANet. (h) Cross-HL. (i) MSRCFNet.

and LiDAR data after feature extraction. However, for C5, C9, and C15, S^{2}ENet and HCT show lower classification accuracy compared to the proposed MSRCFNet. This may be because the proposed MSRCFNet uses a multiscale approach to capture features at different scales, thereby improving the ability of the network to classify complex land cover categories. As shown in Fig. 8(a), a portion of the dataset is affected by clouds, introducing significant interference with classification accuracy. Nevertheless, by observing Fig. 15, the classification map produced by the proposed MSRCFNet exhibits clearer and smoother transitions between categories compared to those generated by other methods. This suggests that the proposed MSRCFNet is still able to capture information between different categories in complex scenes more effectively, further validating the effectiveness and generalizability of the method.

3) *Experiment Result Analysis of MUUFL Dataset:* Table VIII and Fig. 16 present the classification results and full-pixel classification maps of various methods on the MUUFL dataset. The complex terrain and the highly imbalanced sample distribution present a significant challenge for all methods. Nevertheless, the proposed MSRCFNet obtains the highest classification results with OA, AA, and Kappa reaching 89.76%, 79.88%, and 86.41%, respectively. An examination of the classification accuracy for each method reveals that the OAs of certain multisource fusion methods are lower than those of methods using only HSI as input. This may be because the scattered distribution of ground cover and the similarity in elevation information prevent the LiDAR data from providing additional discriminative information. However, MSRCFNet effectively alleviates this issue through its efficient feature fusion strategy, enabling LiDAR data to fully complement HSI, thus surpassing other classification methods in overall performance. Moreover, due to the insufficient number of labeled samples for C9 and C10 and their spatial similarity, most methods struggle to classify both C9 and C10. Nevertheless, the proposed MSRCFNet still demonstrates good classification performance for C9 and C10. This result may result from the effective combination of global and local features through multiscale extraction and cross-modal fusion, which ensures high classification accuracy even with few labeled samples. The proposed MSRCFNet delivers a more precise and visually distinct land cover classification map, achieving closer alignment with the ground truth.

E. Impact of Different Training Samples on The Classification Result

Supervised learning is a data-driven approach that heavily relies on the number of training samples. Therefore, to thoroughly analyze the performance of each algorithm with varying training sample sizes, different training sample percentages {0.5%, 1%, 2%, 3%, 4%, 5%} are used in the experiment. The classification results of each algorithm are

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TABLE VII
CLASSIFICATION RESULT INCLUDING OA (%), AA (%), AND KAPPA WITH STANDARD DEVIATION FOR THE HOUSTON2013 DATASET (OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD)

	HSI Input		HSI and LiDAR Input						
NO.	**HDDA**	**MCFN**	**CoupledCNN**	**S^{2}ENet**	**HCT**	**Sal^{2}RNet**	**MS2CANet**	**Cross-HL**	**Proposed**
C01	79.86±1.45	81.47±2.31	**88.97±3.41**	87.30±2.02	87.51±3.15	85.88±2.81	85.32±1.61	82.18±1.11	87.45±1.61
C02	**89.96±1.78**	83.56±5.22	86.38±5.08	79.00±5.43	85.88±6.63	85.13±5.27	81.42±7.18	87.42±3.62	86.42±7.60
C03	**100.00±0.00**	**100.00±0.00**	99.74±0.46	**100.00±0.00**	**100.00±0.00**	98.38±1.48	97.19±2.83	99.28±0.63	99.25±1.36
C04	91.36±3.46	92.70±4.32	89.30±2.74	**97.30±1.53**	94.88±1.57	95.71±2.06	89.02±2.36	91.81±3.33	95.54±2.90
C05	92.93±0.89	94.82±1.16	90.84±2.62	95.47±2.59	96.89±1.74	94.32±1.72	92.49±1.87	96.24±1.02	**99.77±0.22**
C06	**99.93±0.14**	90.71±10.29	95.57±5.17	98.28±1.32	99.71±0.58	97.18±4.21	99.47±0.38	99.49±0.64	98.48±0.83
C07	85.03±0.94	**88.50±2.66**	88.27±2.98	87.33±2.86	86.06±2.53	80.56±8.07	85.10±2.70	73.65±6.66	87.64±1.28
C08	**89.33±1.58**	87.69±2.87	88.90±5.49	83.98±4.23	86.37±3.25	86.82±6.55	85.68±1.38	79.86±4.88	86.88±2.84
C09	76.19±4.42	81.57±9.90	59.84±3.99	77.24±2.42	79.64±4.57	72.53±6.88	61.89±1.17	65.57±3.97	**82.97±4.68**
C10	90.91±3.26	88.14±3.80	75.60±4.42	**93.24±1.72**	82.25±2.09	86.16±4.32	77.21±2.13	74.87±4.87	87.15±1.64
C11	82.35±2.53	84.70±3.98	76.45±2.47	**94.08±1.55**	90.89±2.49	83.81±4.72	80.62±0.73	71.01±5.49	92.52±1.81
C12	80.43±2.68	80.79±2.62	75.67±3.59	86.30±2.49	83.31±4.64	**86.33±4.28**	82.30±3.50	82.87±2.59	84.46±2.04
C13	88.66±3.20	85.63±7.88	96.71±2.40	87.37±6.36	**97.72±1.72**	90.62±8.51	96.70±2.23	97.37±1.89	94.05±2.58
C14	94.92±4.27	91.89±3.64	97.37±2.22	99.22±1.57	**99.85±0.20**	96.59±3.68	99.60±0.25	95.58±2.06	99.67±0.56
C15	93.90±1.78	98.08±1.27	94.38±1.72	97.45±2.12	95.49±1.97	92.34±2.25	95.22±1.51	93.09±0.87	**98.50±1.62**
OA (%)	87.12±0.88	87.37±1.39	84.10±1.25	89.15±1.12	89.13±0.96	86.82±1.13	84.12±0.99	82.52±1.92	**90.49±1.45**
AA (%)	89.05±0.78	88.68±1.55	86.93±1.07	90.90±1.12	91.10±0.70	88.83±0.72	87.28±0.86	86.02±1.44	**92.05±1.05**
Kappa×100	86.08±0.96	86.34±1.50	82.81±1.35	88.27±1.21	88.25±1.04	85.74±1.22	82.82±1.07	81.08±2.08	**89.72±1.56**

TABLE VIII
CLASSIFICATION RESULT INCLUDING OA (%), AA (%), AND KAPPA WITH STANDARD DEVIATION FOR THE MUUFL DATASET (OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD)

	HSI Input		HSI and LiDAR Input						
NO.	**HDDA**	**MCFN**	**CoupledCNN**	**S^{2}ENet**	**HCT**	**Sal^{2}RNet**	**MS2CANet**	**Cross-HL**	**Proposed**
C01	92.58±1.47	94.53±0.60	**96.40±0.51**	94.85±0.69	90.94±3.11	93.93±2.56	93.68±0.77	86.31±4.52	95.16±0.22
C02	77.92±2.59	78.81±4.13	77.17±3.18	**81.29±2.45**	73.54±3.15	78.97±7.04	76.49±2.87	73.76±9.14	77.82±1.75
C03	**81.48±2.25**	80.15±2.64	77.69±4.91	80.00±0.79	76.82±2.34	72.68±8.15	79.95±2.92	70.05±7.22	80.24±1.65
C04	85.09±3.70	87.9±2.15	83.94±2.10	87.39±1.73	78.91±6.72	82.7±4.31	85.12±4.64	77.26±6.60	**87.99±1.44**
C05	89.06±2.18	**89.55±1.12**	79.37±6.40	89.38±1.06	77.87±7.14	82.56±2.16	88.41±3.13	68.40±9.96	89.08±0.82
C06	**99.20±1.22**	96.88±1.33	85.51±8.91	96.98±0.95	88.32±8.30	87.91±18.19	86.69±9.02	83.79±14.17	97.52±1.26
C07	**84.97±8.87**	80.37±2.01	76.89±5.17	81.30±4.03	63.99±7.81	58.21±3.19	80.66±2.78	63.06±11.05	82.72±3.53
C08	93.48±1.59	94.27±1.17	88.96±4.61	94.22±0.64	93.72±1.31	86.98±5.29	89.81±2.42	83.18±7.38	**94.93±0.99**
C09	62.60±4.34	63.93±2.76	58.20±3.92	70.24±14.47	62.88±8.09	48.10±9.22	61.03±1.97	38.51±7.95	**74.22±3.35**
C10	17.70±19.63	20.00±40.00	**37.55±34.10**	11.83±8.84	16.37±20.32	2.19±4.38	26.17±10.79	2.35±4.70	25.47±5.85
C11	64.40±4.33	82.54±3.36	65.93±3.92	82.74±7.58	**86.97±6.48**	81.90±6.53	66.57±8.10	85.98±7.95	73.55±3.97
OA (%)	88.39±0.30	89.36±0.30	86.66±0.68	89.64±0.73	83.91±1.40	83.74±1.59	87.82±0.40	77.55±1.51	**89.76±0.13**
AA (%)	77.13±1.52	78.99±3.98	75.24±4.24	79.11±2.62	73.67±2.47	70.56±1.59	75.87±0.99	66.60±3.08	**79.88±0.73**
Kappa×100	84.49±0.48	85.87±0.41	82.44±0.84	86.25±0.92	78.51±1.94	78.44±2.37	83.79±0.56	69.55±2.29	**86.41±0.17**

presented in Fig. 17. Typically, a large number of training samples provides sufficient discriminative information, enabling effective training of the supervised learning method, thereby improving the classification accuracy of the algorithm. Experiments reveal that as the quantity of training samples increases, so does the classification accuracy of each method. The classification accuracy of all approaches peaks when the sample size reaches 5%. As illustrated in Fig. 17, the proposed method outperforms other approaches, particularly in scenarios with limited samples. This underscores the effectiveness of integrating multiscale CNN and transformer for classification tasks.

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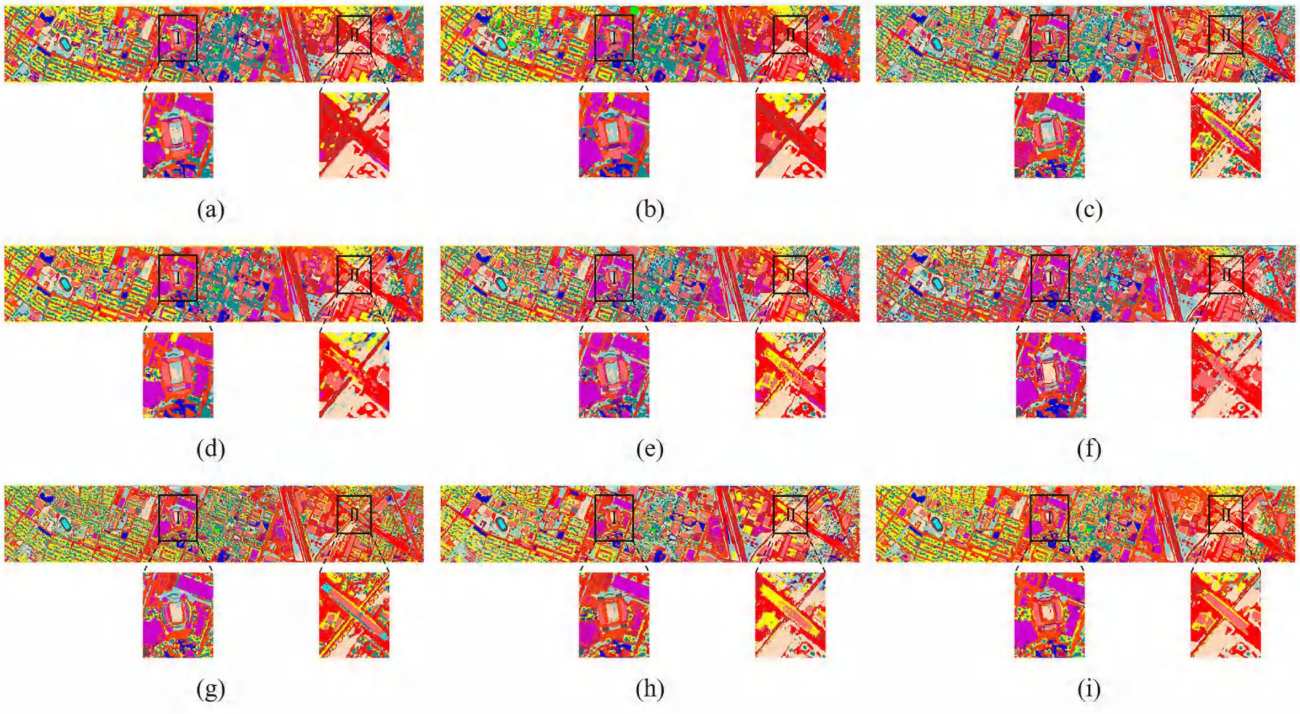


Fig. 15. Full-pixel classification maps for the Houston2013 dataset. (a) HDDA. (b) MCFN. (c) CoupledCNN. (d) S^2ENet. (e) HCT. (f) Sal^2RNet. (g) MS2CANet. (h) Cross-HL. (i) MSRCFNet.

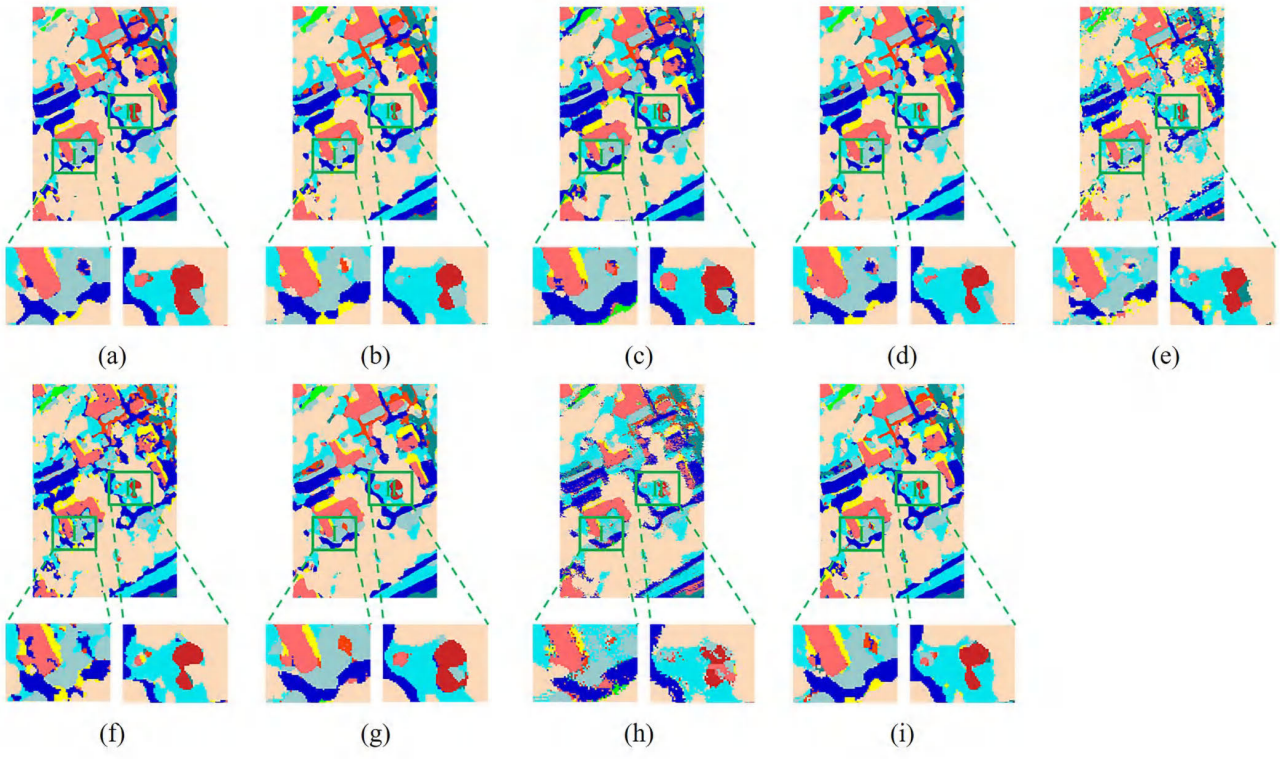


Fig. 16. Full-pixel classification maps for the MUUFL dataset. (a) HDDA. (b) MCFN. (c) CoupledCNN. (d) S^2ENet. (e) HCT. (f) Sal^2RNet. (g) MS2CANet. (h) Cross-HL. (i) MSRCFNet.

F. Ablation Analysis

We conduct ablation experiments in this section on MSRCFNet with different components and discuss the results. The visualization of ablation experiment results intuitively shows how different components affect classification accuracy.

1) *Ablation Analysis of Different Components:* The proposed network consists of four main components: the

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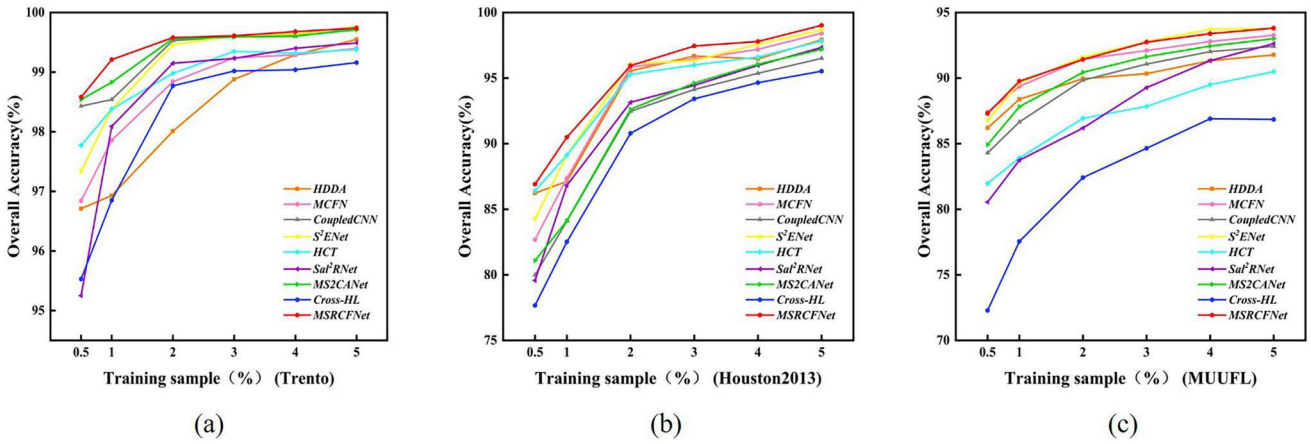


Fig. 17. Classification results of different methods on three datasets with different training samples: (a) Trento dataset; (b) Houston2013 dataset; and (c) MUUFL dataset.

TABLE IX

ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN THE PROPOSED MODEL ON THREE DATASETS (OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD)

	Component				Trento	Houston2013	MUUFL
Cases	**Spe_MHIP**	**Spa_MHIP**	**LiDAR_MHIP**	**CCFM**	**OA (%)**	**OA (%)**	**OA (%)**
model_0	✓	✓			98.02	87.64	89.40
model_1			✓		84.74	39.47	57.28
model_2	✓	✓	✓		99.07	88.12	89.54
model_3				✓	98.58	85.36	85.91
model_4	✓	✓		✓	98.92	90.01	89.48
model_5			✓	✓	98.70	78.96	85.73
model_6	✓	✓	✓	✓	**99.21**	**90.49**	**89.76**

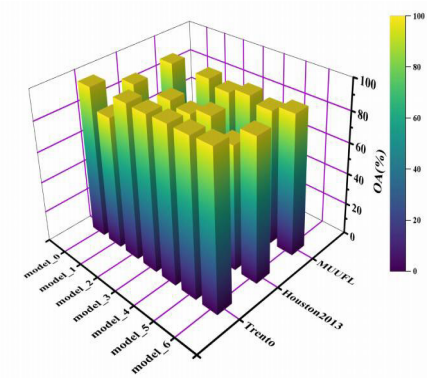


Fig. 18. Ablation results of different components of the proposed model were analyzed and compared on three datasets.

Spe_MHIP module for extracting HSI spectral features, the Spa_MHIP module for extracting HSI spatial features, the LiDAR_MHIP module for extracting elevation features of LiDAR data, and the CCFM for fusing complementary information from heterogeneous data. As shown in Table IX, the model_0 represents a configuration where two MHIP modules are used to extract spectral and spatial features, respectively; the model_1 represents a configuration where only the MHIP module is used to extract elevation features of LiDAR data; the model_2 represents the configuration where the CCFM is removed from the proposed network; the model_3 represents the configuration where only the CCFM is used to fuse heterogeneous data; the model_4 represents the configuration where two MHIP modules are used to extract spectral and spatial features, and the CCFM is used to fuse heterogeneous data; the model_5 represents the configuration where an MHIP module is used to extract elevation features, and the CCFM is used for fusing heterogeneous data; the model_6 represents the complete MSRCFNet. The experimental data and results of the ablation experiment are presented in Table IX and Fig. 18, respectively. Specifically, compared to the model_6, the model_0 reduces the OAs by 1.19%, 2.85%, and 0.26% on the three datasets, respectively. This confirms the critical role of MHIP in LiDAR data feature extraction and the significance of CCFM in heterogeneous data fusion. The CCFM effectively fuses the complementary information of heterogeneous data, thereby significantly improving classification accuracy. The model_1 relies solely on the MHIP module to extract LiDAR data and exhibits suboptimal OAs

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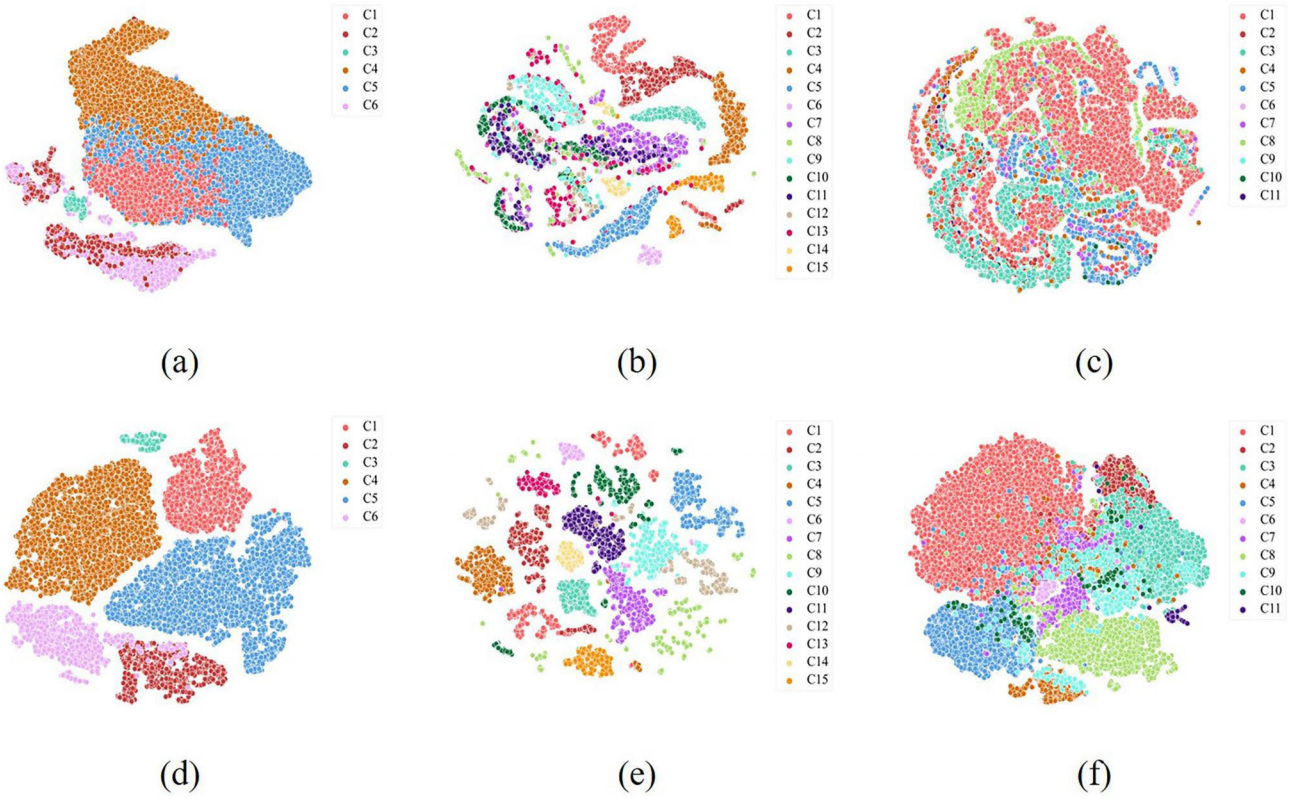
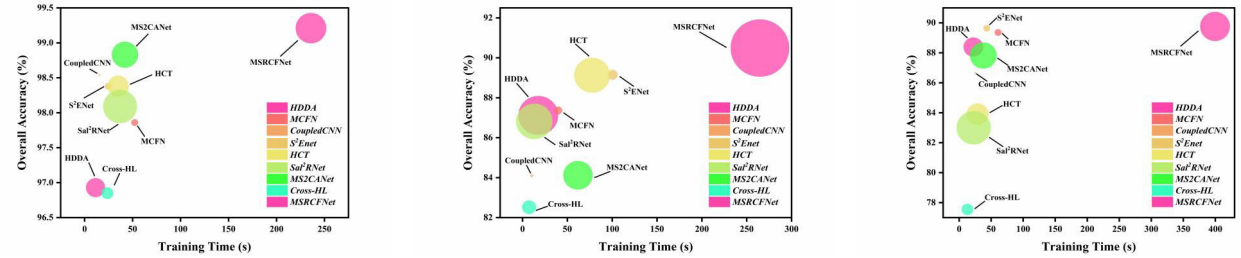


Fig. 19. t-SNE visualization of the proposed method in different datasets. (a) Input data of the Trento dataset. (b) Input data of the Houston2013 dataset. (c) Input data of the MUUFL dataset. (d) Output data of the Trento dataset. (e) Output data of the Houston2013 dataset. (f) Output data of the MUUFL dataset.

performance on the three datasets. This demonstrates that without HSI data, extracting critical information solely using the MHIP module for feature extraction from LiDAR data is challenging. In the case of the model_2, extracting spectral, spatial, and elevation features separately solely using the MHIP module results in an OA lower than that of model_6 across all three datasets. This result further underscores the importance of CCFM. The model_3 does not extract features but directly inputs raw data into the CCFM for feature fusion. Compared to the model_6, the OA of each dataset is reduced by 0.63%, 5.13%, and 3.85%, respectively. This indicates that CCFM cannot effectively integrate multimodal raw data without sufficient feature input and emphasizes the importance of using the MHIP module for multiscale feature extraction. In addition, the Houston2013 dataset and MUUFL dataset contain more complex ground categories, where different categories exhibit relatively similar spectral characteristics. Therefore, directly fusing multimodal data without sufficient feature extraction results in a significant decrease in OA. In contrast, model_4, which integrates both feature extraction and fusion of HSI, achieves higher classification accuracy compared to model_0, which is solely dedicated to HSI feature extraction. The model_5, which combines feature extraction and fusion for LiDAR data, achieves superior classification accuracy compared to model_1, which only performs feature extraction for LiDAR data. This shows that while the feature extraction of HSI and LiDAR data individually captures critical information from each dataset, the feature fusion through CCFM further enhances the discriminative ability of the network. The ablation results confirm that each component contributes to the overall performance of MSRCFNet, with the combination of all components achieving optimal classification performance.

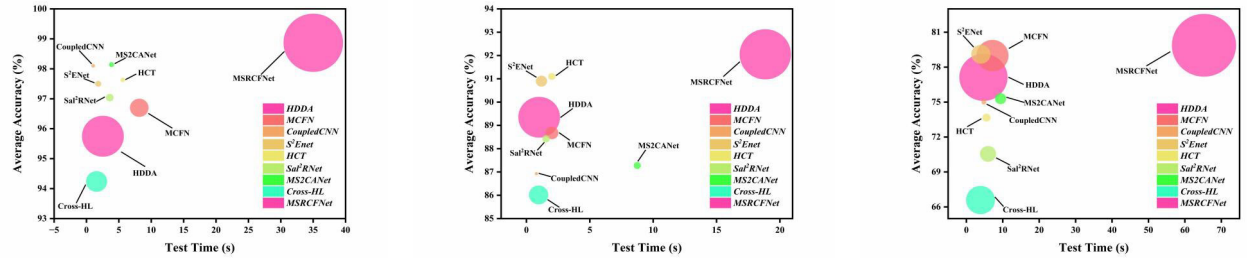
2) *t-SNE Visual Evaluation and Analysis:* The t-distributed stochastic neighbor embedding (t-SNE) [72] is employed to visualize the distribution of the original data and the data output by MSRCFNet in a low-dimensional feature space, thereby providing an intuitive evaluation of the feature extraction capability of the proposed network. As shown in Fig. 19, different colors represent the different label categories. Fig. 19(a)–(c) represents the feature distribution of the original data across the three datasets, while Fig. 19(d)–(f) represents the feature distribution of the data after MSRCFNet output on the three datasets. The features of the original data overlap chaotically. Examples include C1 and C5, C2, and C6 in the Trento dataset, C8, C10, C11, and C13 in the Houston2013 dataset, and C1, C6, and C7 in the MUUFL dataset. The proposed MSRCFNet serves as a feature mapping method that transforms hard-to-distinguish features into easily distinguishable ones. As shown in Fig. 19(d)–(f),

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(a) (b) (c)

Fig. 20. Comparison of OA, parameter quantities, and average training time across three datasets. Larger bubble sizes reflect greater parameter quantities. (a) Trento dataset. (b) Houston2013 dataset. (c) MUUFL dataset.



(a) (b) (c)

Fig. 21. Comparison of AA, FLOPs, and average test time across three datasets. Larger bubble sizes reflect greater FLOPs. (a) Trento dataset. (b) Houston2013 dataset. (c) MUUFL dataset.

after the proposed MSRCFNet processing, the same categories become more compact, and different categories are clearly distinguished. It is worth noting that, due to the imbalanced sample distribution in the MUUFL dataset, the original data exhibits numerous misclassifications. However, the multiscale feature extraction and fusion in the proposed MSRCFNet disperse the data features, thereby enhancing category discrimination.

G. Model Complexity Analysis

In order to objectively evaluate the computational efficiency of the proposed MSRCFNet and the other eight comparative algorithms, we refer to [73] to conduct complexity comparison experiments. In the complexity comparison experiments, parameter quantities (Params) and floating-point operations per second (FLOPs) are used as evaluation metrics to comprehensively assess the performance of each algorithm. During the complexity comparison experiments for each model, a batch size of one is utilized when the network is fed with each dataset.

By observing Figs. 20 and 21, it can be found that although CoupledCNN shows the lowest Params and FLOPs among all the datasets, its simple network architecture hinders its ability to capture complex features, thus limiting its classification performance. Similarly, HCT and Sal^{2}RNet also have a similar situation. Especially for Sal^{2}RNet, the Params of this network on the Trento and MUUFL datasets are higher than those of the proposed MSRCFNet, but the OA and AA of Sal^{2}RNet on the three datasets are lower than the OA and AA of MSRCFNet. HDDA, which only uses HSI as the input, exhibits higher complexity, which may be caused by the large number of convolutional layers and pooling layers used in this network. Nevertheless, the classification performance of this algorithm on the Trento dataset and the Houston2013 dataset still has no competitive advantage compared with the proposed MSRCFNet. Similarly, Cross-HL, as a network based on the transformer architecture, achieves the worst classification performance on the three datasets while maintaining relatively high Params and FLOPs. In contrast, although the proposed MSRCFNet has a higher computational complexity than most of the comparative algorithms, it achieves the best OA and AA on the three datasets. This is attributed to the effective integration of the local feature extraction ability of the CNN and the global modeling ability of the attention mechanism, constructing a network architecture that can fully extract multiscale features and effectively fuse heterogeneous data. Despite the increase in computational complexity, this tradeoff significantly improves the discriminative ability of the model for the subtle differences and complex land cover in remote sensing images, enabling the model to maintain stable and accurate classification results in complex scenarios.

V. Conclusion

In this article, an MSRCFNet based on CNN and attention mechanisms is proposed for hyperspectral and LiDAR land cover classification. The proposed network consists of three parallel MHIP modules, a CCFM, and a classification module. The MHIP module is used to extract multiscale spatial, spectral, and elevation features. Different attention mechanisms are adapted in the CCFM to perform adaptive fusion of the

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complementary features from multisource remote sensing data. The classification module maps the semantic features to the final classification results. Comparisons with various classification techniques on three publicly available HSI-LiDAR datasets demonstrate that the proposed method consistently achieves higher classification accuracy. Ablation experiments demonstrate the effectiveness of both the MHIP module and CCFM in the proposed MSRCFNet. In addition, the impact of hyperparameters on network performance is discussed, and an analysis of data feature visualization is conducted. In future research, we aim to improve the computational efficiency of the model while ensuring its classification performance for multisource remote sensing data.

REFERENCES

[1] C. Toth and G. Józków, "Remote sensing platforms and sensors: A survey," *ISPRS J. Photogramm. Remote Sens.*, vol. 115, pp. 22–36, May 2016.

[2] W. Cao, L. Dong, L. Wu, and Y. Liu, "Quantifying urban areas with multi-source data based on percolation theory," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111730.

[3] Y. Cheng, K. Zhou, J. Wang, and J. Yan, "Big Earth observation data integration in remote sensing based on a distributed spatial framework," *Remote Sens.*, vol. 12, no. 6, p. 972, Mar. 2020.

[4] M. Appel, F. Lahn, W. Buytaert, and E. Pebesma, "Open and scalable analytics of large Earth observation datasets: From scenes to multidimensional arrays using SciDB and GDAL," *ISPRS J. Photogramm. Remote Sens.*, vol. 138, pp. 47–56, Apr. 2018.

[5] X. Zhu, F. Cai, J. Tian, and T. Williams, "Spatiotemporal fusion of multisource remote sensing data: Literature survey, taxonomy, principles, applications, and future directions," *Remote Sens.*, vol. 10, no. 4, p. 527, Mar. 2018.

[6] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[7] R. Ribeiro, G. Cruz, J. Matos, and A. Bernardino, "A data set for airborne maritime surveillance environments," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 9, pp. 2720–2732, Sep. 2019.

[8] E. A. Cloutis, "Review article hyperspectral geological remote sensing: Evaluation of analytical techniques," *Int. J. Remote Sens.*, vol. 17, no. 12, pp. 2215–2242, Aug. 1996.

[9] A. Sharifi, "Estimation of biophysical parameters in wheat crops in Golestan province using ultra-high resolution images," *Remote Sens. Lett.*, vol. 9, no. 6, pp. 559–568, Mar. 2018.

[10] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Dec. 2021.

[11] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[12] G. Rivera, R. Porras, R. Florencia, and J. P. Sánchez-Solís, "LiDAR applications in precision agriculture for cultivating crops: A review of recent advances," *Comput. Electron. Agricult.*, vol. 207, Mar. 2023, Art. no. 107737.

[13] Z. Wang and M. Menenti, "Challenges and opportunities in LiDAR remote sensing," *Frontiers Remote Sens.*, vol. 2, Mar. 2021, Art. no. 641723.

[14] Z. Zhang, T. Li, X. Tang, X. Lei, and Y. Peng, "Introducing improved transformer to land cover classification using multispectral LiDAR point clouds," *Remote Sens.*, vol. 14, no. 15, p. 3808, Aug. 2022.

[15] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.

[16] S. E. Reutebuch, H.-E. Andersen, and R. J. McGaughey, "Light detection and ranging (LiDAR): An emerging tool for multiple resource inventory," *J. Forestry*, vol. 103, no. 6, pp. 286–292, Sep. 2005.

[17] M. Schmitt and X. X. Zhu, "Data fusion and remote sensing: An evergrowing relationship," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 4, pp. 6–23, Dec. 2016.

[18] M. Dalponte, L. Bruzzone, and D. Gianelle, "Fusion of hyperspectral and LiDAR remote sensing data for classification of complex forest areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1416–1427, May 2008.

[19] Q. Liu, D. Xue, Y. Tang, Y. Zhao, J. Ren, and H. Sun, "PSSA: PCA-domain superpixelwise singular spectral analysis for unsupervised hyperspectral image classification," *Remote Sens.*, vol. 15, no. 4, p. 890, Feb. 2023.

[20] W. Li, S. Prasad, J. E. Fowler, and L. M. Bruce, "Locality-preserving discriminant analysis in kernel-induced feature spaces for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 5, pp. 894–898, Sep. 2011.

[21] F. M. B. Van Coillie et al., "Optimized feature fusion of LiDAR and hyperspectral data for tree species mapping in closed forest canopies," in *Proc. 7th Workshop Hyperspectral Image Signal Processing, Evol. Remote Sens. (WHISPERS)*, Jun. 2015, pp. 1–4.

[22] B. Bigdeli, F. Samadzadegan, and P. Reinartz, "A decision fusion method based on multiple support vector machine system for fusion of hyperspectral and LiDAR data," *Int. J. Image Data Fusion*, vol. 5, no. 3, pp. 196–209, Jun. 2014.

[23] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.

[24] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 3011–3024, Jun. 2017.

[25] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Aug. 2016.

[26] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[27] S. K. Roy, A. Deria, D. Hong, M. Ahmad, A. Plaza, and J. Chanussot, "Hyperspectral and LiDAR data classification using joint CNNs and morphological feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530416.

[28] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506812.

[29] H. Pan, Y. Zhu, H. Ge, M. Liu, and C. Shi, "Multiscale cross-fusion network for hyperspectral image classification," *Egyptian J. Remote Sens. Space Sci.*, vol. 26, no. 3, pp. 839–850, Dec. 2023.

[30] Y. Fan et al., "MSLAENet: Multiscale learning and attention enhancement network for fusion classification of hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 10041–10054, 2022.

[31] H. Gao, H. Feng, Y. Zhang, S. Xu, and B. Zhang, "AMSSE-net: Adaptive multiscale spatial–spectral enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531317.

[32] X. Wang, J. Zhu, Y. Feng, and L. Wang, "MS2CANet: Multi-scale spatial–spectral cross-modal attention network for hyperspectral image and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–17, 2024.

[33] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[34] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[35] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.

[36] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[37] S. K. Roy, A. Sukul, A. Jamali, J. M. Haut, and P. Ghamisi, "Cross hyperspectral and LiDAR attention transformer: An extended self-attention for land use and land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512815.

[38] Y. Zhang, Y. Peng, B. Tu, and Y. Liu, "Local information interaction transformer for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1130–1143, 2023.

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[39] Y. Zhang et al., "Multimodal transformer network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514317.
[40] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.
[41] L. Sun, X. Wang, Y. Zheng, Z. Wu, and L. Fu, "Multiscale 3D–2D mixed CNN and lightweight attention-free transformer for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 2100116.
[42] W. Wang et al., "Dual-branch feature fusion network based cross-modal enhanced CNN and transformer for hyperspectral and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[43] M. Wang, Y. Sun, J. Xiang, and Y. Zhong, "CITNet: Convolution interaction transformer network for hyperspectral and LiDAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535918.
[44] S. Morchhale, V. P. Pauca, R. J. Plemmons, and T. C. Torgersen, "Classification of pixel-level fused hyperspectral and LiDAR data using deep convolutional neural networks," in *Proc. 8th Workshop Hyperspectral Image Signal Processing, Evol. Remote Sens. (WHISPERS)*, Aug. 2016, pp. 1–5.
[45] C. Ge, Q. Du, W. Sun, K. Wang, J. Li, and Y. Li, "Deep residual network-based fusion framework for hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2458–2472, 2021.
[46] H. Wang, X. Liu, Z. Qiao, G. Wang, and H. Chen, "Multimodal remote sensing data classification based on Gaussian mixture variational dynamic fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5621214.
[47] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.
[48] H. Zhang, J. Yao, L. Ni, L. Gao, and M. Huang, "Multimodal attention-aware convolutional neural networks for classification of hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3635–3644, 2023.
[49] C. Lee and E. Choi, "Optimizing feature extraction for multiclass problems," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 521–528, Mar. 2001.
[50] J. Hu, B. Tu, Q. Ren, X. Liao, Z. Cao, and A. Plaza, "Hyperspectral image classification via multi-scale multi-angle attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, 2024.
[51] H. Ge et al., "Cross attention-based multi-scale convolutional fusion network for hyperspectral and LiDAR joint classification," *Remote Sens.*, vol. 16, no. 21, p. 4073, Oct. 2024.
[52] Y. Tong et al., "Multi-scale feature extraction and total variation based fusion method for HSI and LiDAR data classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 5433–5436.
[53] Y. Xi and Z. Ye, "Multi-scale feature extraction for joint classification of hyperspectral and LiDAR data," *J. Beijing Inst. Technol.*, vol. 32, no. 1, pp. 13–22, Feb. 2023.
[54] Q. Meng, M. Zhao, L. Zhang, W. Shi, C. Su, and L. Bruzzone, "Multilayer feature fusion network with spatial attention and gated mechanism for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[55] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020.
[56] H. Liu, F. Liu, X. Fan, and D. Huang, "Polarized self-attention: Towards high-quality pixel-wise mapping," *Neurocomputing*, vol. 506, pp. 158–167, Sep. 2022.
[57] X. Meng, L. Zhu, Y. Han, and H. Zhang, "We need to communicate: Communicating attention network for semantic segmentation of high-resolution remote sensing images," *Remote Sens.*, vol. 15, no. 14, p. 3619, Jul. 2023.
[58] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, no. 1, 2015, Art. no. 258619.
[59] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.
[60] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.
[61] C.-F. R. Chen, Q. Fan, and R. Panda, "CrossViT: Cross-attention multi-scale vision transformer for image classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Aug. 2021, pp. 357–366.
[62] X. Chen, X. Wang, J. Zhou, and C. Dong, "Activating more pixels in image super-resolution transformer," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 22367–22377.
[63] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.
[64] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[65] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.
[66] X. Du and A. Zare, "Technical report: Scene label ground truth map for MUUFL Gulfport data set," Univ. Florida, Gainesville, FL, USA, Tech. Rep., 2017.
[67] P. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, "MUUFL gulfport hyperspectral and LiDAR airborne data set," Univ. Florida, Gainesville, FL, USA, Tech. Rep. REP-2013-570, 2013.
[68] J. Zhao, L. Hu, Y. Dong, and L. Huang, "Hybrid dense network with dual attention for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 23, p. 4921, Dec. 2021.
[69] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.
[70] S. Fang, K. Li, and Z. Li, "S^2ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[71] J. Li, Y. Liu, R. Song, Y. Li, K. Han, and Q. Du, "Sal^2RN: A spatial–spectral salient reinforcement network for hyperspectral and LiDAR data fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500114.
[72] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, Nov. 2008.
[73] Y. Zhao, W. Bao, J. Xu, and X. Xu, "BIHAF-net: Bilateral interactive hierarchical adaptive fusion network for collaborative classification of hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 15971–15988, 2024.



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